Nonlinear Wave and Dispersive PDE

— ***EYNTKA*** **Series** —

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Contents

Summary

It as assumed that the reader is comfortable with the concept introduced in a first course in analysis and linear algebra, importantly that smooth functions and vector-spaces are comfortable concepts.

Introduction Differential Equations

To avoid super-script counting, the book shall work with smooth functions (or else the order of differentiability must be kept track). This shall not change the theory for ODE's and PDEs, especially since later it shall be shown that even working with non-smooth differentiable functions shall end up having the same theory (due to the advent of Distribution Theory [1])

Let $f : \mathbb{R} \to \mathbb{R}$. When working smooth functions, we may take the derivative of the function. This gives the following function taking smooth functions and returning smooth functions:

$$f \mapsto f'$$

Due to the fundamental theorem of calculus, the inverse of this procedure is called *integration*, and gives a unique inverse up to a constant:

$$f \mapsto F + c \qquad c \in \mathbb{R}$$

where F may be defined as $F(x) = \int_t^x f(x)dx$ for appropriate t. Using this result, it is possible to find the smooth function that satisfies an arbitrary degree of differentiation, for example:

$$f^{(n)} = 0$$

which is a polynomial of degree n, or more generally

$$f^{(n)} = g \qquad \text{equiv.} \qquad f^{(n)} - g = 0$$

for some smooth functions g[2].

Given this simple scenario, the question may arise if we may solve more complicated algebraic expression. For example, does there exist a solution to any of these:

$$f = f' \qquad f' + f = 0 \qquad f'' + f' + f = 0$$

[1] see chapter 3

[2] Note that the solution may not always come in an elementary form, that is expressible as a polynomial-combination and composition of x, log, e^x, or a trigonometric identity

More precisely, we ask if any arbitrary algebraic expression on a smooth function f combined by using the 4 usual arithmetic operators $(+, -, \times, \mathrm{div})$ along with differentiation yields a solution:

$$p(f) = 0$$

more commonly written as:

$$F(x, f(x), f'(x), ..., f^{(n)}(x)) = 0$$

The study of the solution to such equations is called the study of *Ordinary Differential Equations* (ODEs). If we have a smooth function of the form $f : \mathbb{R} \to \mathbb{R}^m$, this is still the study of ODEs, where it is solving a *system of ODEs*. When working with multi-variable functions and partial derivatives, the study of the solution to such equations is called the study of *Partial Differential Equations* (PDEs). The study of ODEs can be thought of as a special case of the study of PDEs.

This book shall start with the study of ODEs in the first chapter, and proceed with the rest of the chapter with the study of PDEs. Only one chapter is dedicated to ODEs as they are a solved problem where most ODEs have a general solution that can be found algorithmically, while this is provably not the case for most PDEs.

From a more physical perspective, differential equations will give you how certain parameters relate to each, for PDEs it is usually over time. As a simple example, take:

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial x} \qquad \text{or} \qquad \frac{\partial}{\partial t} - \frac{\partial}{\partial x} = 0 \tag{1}$$

What this equation is giving us is a restriction we require a function $f(t, x)$ to obey, that is:

$$\frac{\partial f}{\partial t} - \frac{\partial f}{\partial x} = 0$$

For example $f(x, t) = x + t$ would satisfy this equation. One key skill to learn when solving problems in PDEs is to be able to "read" such an equation. For example, equation (1) shows that the speed at which our "system" is evolving over time is identically propositional to the speed at which our "system" is moving. Sometimes, it is quite subtle what an equation may represent. For example:

$$\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} = 0$$

is very similar to equation (1), however it represents how a *wave* evolves over time, with behavior that is very different from equation (1).

The reason why finding solutions to PDEs is interesting to me is because the derivative can give lot's of local information about a system, and sometimes that's all the information we have. For example, it is much easier to describe the force of gravity as

$$F = ma$$

where $a = dv/dt$, and $v = d(\text{displacement})/dt = dD/dt$. We can use this equation to construct a gravitational field to how any object at any point will have a certain force on them represented by a vector pointing downwards times the mass of the object, but this "global solution" has a lot more information than is needed. More generally, it is often easier to come up with the "localizations", since they are easier to define or control most of the time.

Also, after seeing how heat equations and diffusion equations are described I have a new motivation: I wonder if at some point these techniques of relating "local information" in describing how it's dynamics evolve (in physics, it would be how it changes in time, in more general scenarios it can be a recursive function or a sequence) will at some point lend itself useful in solving more general problems. For example, let's say we have a particularly complicated mathematical object. Let's say we can solve it in some nice special cases, perhaps with some restricting assumptions. Can we then step-by-step evolve the problem (or model an evolution of the problem) to eliminate the restrictions? Can we localize in some way to simplify it, and glue that information together? These feel like modelling something like a PDE.

As a final motivator, every differential form has a "canonical representative" in that there always exists representative ω such that ω vanishes under the Laplace an: $\Delta\omega = 0$ (functions, and more generally forms, that vanishes under the Laplace an are called *harmonic*). This is the beginning of *Hodge Theory*, which is central in studying the cohomology groups of smooth manifolds. In the same vein as geometry but from another perspective, PDEs are also used to model *Ricci flow*, which is about the flow of a manifold given certain parameters as it changes across time.

PDEs

Partial differential equations, often abbreviated as PDEs, let's us *relate* quantities. Another way of seeing how PDEs are different is that ODE's usually only model one variable (ex. angle of a pendulum), while a PDE can angle a whole set of points at once (ex. all the temperatures of a rod, as they change over time). For example, we may hope that the position of a particle (locally) relates to the force put on that particle (locally), and we want to do this for every particle in a system (more generally a continuum). For a more mathematical motivation, they show up more and more often in differential geometry (think of lee bracket's, lie derivative, and geometric analysis), they let us study D-modules (modules over the ring of differential operators, which tries to find aglebraic properties of differential operators), and geometry topology (PDEs were important in defining Ricci flow to show that any loop on a locally $\mathbb{R}^3$ space which can contract to a point is in fact a 3-sphere).

There is no unified study of PDEs, instead there is an amalgamation of "models" or "problems" that were of interest that are represented as a PDE. In the most abstract form, PDEs are smooth functions $f : U \to \mathbb{R}$ for a set $U \subseteq \mathbb{R}^n$ (usually with some extra conditions like being compact or open) solving a function:

$$F\left(\vec{x}, f(\vec{x}), \frac{\partial f}{\partial x_1}, ...\right) = 0 \tag{2}$$

where F takes takes all possible partials of f. Restrictions on F are usually that it is a polynomials combination or a trigonometric combination, however this is not necessarily the case. It is often the case that the solution function is denoted by u. Example of PDEs include:

- transport equation: Models how something move's across time:

$$u_t + cy_x = 0$$

- wave equation: Models how a "wave" moves across time. Exapmles of wave are "literal" water waves, gravitaional waves, sound waves, physical waves (ex. a string or membrane vibrating), and so forth. What characterizes a wave is that the more "curved" any part of the wave is (i.e. the more concave it is), the more it accelerrates. To capture the curve-ness, the term $\frac{\partial^2}{\partial x^2}$ is used (which you can think of as representing the average), while the acceleration is captured by $\frac{\partial^2}{\partial t^2}$. Hence put together, we get:

$$\frac{\partial^2}{\partial t^2} = \frac{\partial^2}{\partial x^2} \quad \Leftrightarrow \quad \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} = 0$$

if we add more dimensions, we would write:

$$\frac{\partial^2}{\partial t^2} - \sum_i^n \frac{\partial^2}{\partial x_i^2} = \frac{\partial^2}{\partial t^2} - \nabla = 0$$

Due to the proclivity of wave equations, a square symbol is often used to denote the wave operator:

$$\square = \frac{\partial^2}{\partial t^2} - \nabla$$

Hence, a function $u(t, \mathbf{x})$ would represent a [homogeneous] wave equation if it satisfies:

$$\square u = 0$$

- heat equation: The heat equation, usually written (in 1 dimension) as $\frac{\partial}{\partial t} = \frac{\partial^2}{\partial x^2}$ represents how heat naturally averages over time. In higher dimensions, this is often written as:

  $$u_t = k\Delta u$$

  for some constant k and the *Laplace operator*: $\Delta u = \sum_i \frac{\partial u}{\partial^2 x}$

- Maxwell's equation: These are a more complicated version of the wave equations to take into account some interaction of the electromagnetic force:

  $$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$
  $$\nabla \cdot \vec{B} = 0$$
  $$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$
  $$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

  where E and B represent the *electric* and *magnetic* fields.

- Other famous equations note remembering are: Schrödinger's equation, Navier-Stokes equation, Dirac Equations, Yang-Mills equations, Einsteins General Relativity equations, and the Black-Scholes equation.

As it stands, the set of solutions for such equations can be more general than we want. For example, when solving a heat equation or a wave equation, a moment in time and a finite area is usually considered. Thus, when solving a PDE, it usually comes with a set of questions and restrictions. This set along with the equation is usually called a *Problem*. The two most types of problems considered in this book are:

1. *Initial Value Problem (IVP)*: One of the variables is considered a variable of time, and we usually have an *initial state* at a certain specified time t_0:

$$u|_{t_0} = u_0$$

for example, if u is modelling the temperature in a rod, then at time t_0, half the rod may be at 100 degrees, while the other half at 0 degrees. This information would be captured in u_0.

2. *Boundary Value Problem*: If U is the domain of f, then a known function is imposed on this domain:

$$u|_{\partial U} = \varphi$$

where φ is a known (smooth) function defined on ∂U.

These two can also be combined, the resulting problem is sometimes called a *mixed problem* due to the prevalence of combining these two types of problems. When finding a solution to a problem, we often would want the function (or functions in the case of a system of PDEs) to satisfy the following:

1. *Existence*: Given the restrictions (ex. choices of boundary conditions or choices of time), there exists a solution.

2. *Uniqueness*: Given a specific choice of restrictions, there exists a unique solution

3. *Continuity*: Given small variations of the choices of conditions, there still exists a solution that varies continuously (i.e. that the set of solutions forms an open set[3])

What "continuously" means here can depend: It may be that the desired solution varies continuously, smoothly, or any condition in between. A problem that satisfies all three of the above conditions is called a *well-posed* problem. Most problems that will be studied will be well-posed, however there will exist *ill-posed* problems when considering a type of problem called the *inverse problem* later in this book.

Classification of PDEs

As we saw, PDE's are very general and are not well suited for classifications. However, there are some broad categories that they may be put into that helps with understanding how to approach them. A *linear* PDE is one where F in equation 2 is a linear transformation:

$$L\left(\vec{x}, f(\vec{x}), \frac{\partial f}{\partial x_1}, ...\right) = g(x) \tag{3}$$

if $g(x) \equiv 0$, then the above is called a *homogeneous* linear PDE. If $g(x) \neq 0$, then it is called an *inhomogeneous* PDE. Note that the coefficients of the linear combination may be *functions* (consider this a $C^\infty(U)$-module). Linear PDEs are some of the simplest types as they result in some simple analysis solutions through the convergence of the Fourier series as we shall see. Non-linear PDEs are a more complicated story. As linear PDEs are a solved problem, the limits of linearity where pushed to try and extend the techniques as much as possible.

[3]In the natural topology imposed on this function space, see a course in functional analysis or an advanced analysis course for more details

Linearity can be extended can generalised to try to solve a more broad class of problems. *Semi-linear* equations are those whose highest order coefficient(s) will appear as linear terms, the coefficients being functions of free variables (i.e. as part of the dimension of the space of the problem). For example, a general 2nd order semi-linear PDE in two variables[4] would be:

$$a_1(x,y)u_{xx} + a_2(x,y)u_{xx} + a_3(x,y)u_{xx} + a_4(x,y)u_{xx} + f(u_x, u_y, x, y, u) = 0$$

Note that the lower-order terms have no linearity restrictions, and hence are put within the general function f. A *quasi-linear* PDE is an extension of this concept, were lower order terms are allowed as well:

$$a_1(u_x,u_y,x,y,u)u_{xx}+a_2(u_x,u_y,x,y,u)u_{xx}+a_3(u_x,u_y,x,y,u)u_{xx}+a_4(u_x,u_y,x,y,u)u_{xx}+f(u_x,u_y,x,y,u) = 0$$

If f is linear, then the equation is called a *linear equation.* Most equations we shall be working with will in fact be linear, for example most wave and heat equations will be linear equations.

Linear equations shall over plenty of important PDEs. Consider the following:

here

[4]Do not get confuse dimension of the problem with the order of the PDE. There *can* be more or less dimensions, hence variables!

1

1-dimensional PDEs

The goal of this chapter is to be able to solve basic homogeneous wave equations of the form $\Box u = 0$. This will build up bit by bit, form the simplest case of a one-dimensional standing wave in a "perfect" system, increasingly more complex and "realistic" situations.

1.1 1st order

Consider:

$$au_t + bu_x = 0$$

where a and b are smooth functions. Physically, the left hand side may be interpreted as the derivative of the vector-field $\ell = (a, b)$. The integral lines of this field (lines representing the "flow")would be of the form:

$$\frac{dt}{a} = \frac{dx}{b}$$

Constant variables

Starting with the toy-case of a, b, being constant, then we have the integral liens:

$$\frac{t}{a} - \frac{x}{b} = C$$

for some constant C. This makes solving for u particularity easy: given any smooth function φ:

$$u = \varphi\left(\frac{t}{a} - \frac{x}{b}\right)$$

This is considered the *general solution*. If we consider a IVP where $u|_{t=0} = f$. Then

$$\varphi(-x/b) = f(x) \quad \Rightarrow \varphi(x) = f(-bx)$$

giving us:

$$u = f(x - ct) \qquad c = b/a$$

which solves:

$$\begin{cases} au_t + bu_x = 0 \\ u(x,0) = f(x) \end{cases}$$

such a solution is called a *running wave*, due to the physical interpretation of this equation.

Let us now develop for more general cases. Some care must now be taken to insure our problem is *well-posed*

Example 1.1.1: Ill-posed Problem

Consider $u_t + tu_x = 0$. This equation has integral curves

$$tdt - dx = 0$$

which, solving by using the method of separable variables, we get:

$$x - \frac{1}{2}t^2 = C$$

Hence, we see that $u = \varphi(x - 1/2t^2)$ is a solution to this PDE. Notice that the input must be symmetrical across the t axis. This shall give some important constraints for possible IVP.

Consider this equation with IVP $\varphi(0 - 1/2t^2) = u(0,t) = g(t)$, that is we want the wave to follow the path $g(t)$ at point $x = 0$. Namely, we have $\varphi(-1/2t^2) = g(t)$, meaning t *must* be an even function with respect to t, or this equation is not solvable. Notice that with a badly picked g (ex. g is odd with respect to t), this is not possible. Furthermore, even if we limit to even function or functions where $t > 0$, then there is no restriction on all points $x > 1/2t^2$, making us lack uniqueness for the solution.

Hence, more care must now be put with choosing our IVP. That being said, let us also consider the non-homogeneous case:

$$au_t + bu_x + f(x,t,u) = 0 \qquad \Leftrightarrow \qquad au_t + bu_x = f(x,t,u)$$

where the negative is incorporated into f on the right hand side. Then the integral curves follow:

$$du = u_t dt + u_x dx = (au_t + bu_x)\frac{dt}{a} = f\frac{dt}{a}$$

Which reduces our problem to:

$$\frac{dt}{a} = \frac{dx}{b} = \frac{du}{f}$$

Example 1.1.2: Non-homogeneous Solutions

1. Take $u_t + u_x = x$. This gives
$$dx = dt = \frac{du}{x}$$
Solving two pairs, we get $x - t = C$ and $u - 1/2x^2 = D$. This time, there is a u as part of our equations. Hence we know we must satisfy the integral lines
$$u = \frac{1}{2}x^2 + D$$
If there were no further restrictions, we can make D an arbitrary smooth function, however as $x - t = C$, we may use the same trick as before and define:
$$u = \frac{1}{2x}^2 + \varphi(x - y)$$
Which is the general solution. Show that if the IVP is $u|_{t=0} = 0$ the solution is
$$u = xt - \frac{1}{2}t^2$$
and verify that this satisfies $u_x + u_t = x$
2. Take $u_t + xu_x = xt$. This gives:
$$\frac{dt}{1} = \frac{dx}{x} = \frac{du}{xt}$$
(bunch more computation, can verify on page 23, answer: $u = x(t - 1) + \varphi(xe^{-t})$).

Semi-Linear

Let us now consider the case where $a = a(x, t)$ and $b = (x, t)$ so that we are working with semi-linear 1st order equations. We it is also the case that $f = c(x, t)u + g(x, y)$ (that is, f is a linear combination of u with a constant[1]), then the equation is called linear.

Example 1.1.3: Semi-Liner and Linear equations

1. Take $u_t + xu_x = u$ so that the integral curves are:
$$\frac{dt}{1} = \frac{dx}{x} = \frac{du}{u}$$
Taking the first two equations in this equality, we can solve it using the method of variable separation to get:
$$x = Ce^t$$
Taking the first and third, we get:
$$\frac{du}{u} = dt \Rightarrow u = De^t$$

[1] Recall our constants are usually smooth functions

as $C = xe^{-t}$ is a constant, we get:

$$u = \varphi(xe^{-t})e^t$$

form which we may impose an IVP problem, for example if $u|_{t=0} = x^2$ we would get $u = x^2e^{-t}$.

2. Consider

Quasi-Linear

In the 1-dimensional first order case, this means that the coefficients of u_t and u_x now also permit u as a free-variable. The general equation for integral curves becomes:

$$\frac{dt}{a} = \frac{dx}{b} = \frac{du}{f}$$

These may present more problems when looking for solutions:

Example 1.1.4: Quasi-linear Trouble
Consider:

$$u_t + uu_x = 0$$

Giving

$$\frac{dt}{1} = \frac{dx}{u} = \frac{du}{0}$$

The last equation is a shorthand for saying that u is constant: $u = C$. From the other equation, we get:

$$ut - x = D \tag{1.1}$$

We now have two equations with constants, which is enough to piece together the general solution, namely since C, D are the constants we get that $u = C = \varphi(D) = \varphi(ut - x)$:

$$u = \varphi(ut - x)$$

However, note that this is defined *implicitly* (notice the repetition of u). This shows that we may find a solution if and only if

$$\frac{\partial}{\partial u}(u - \varphi(ut - x)) = 1 + \varphi'(ut - x)t \neq 0$$

which with some manipulation shows us is possible for all $t > 0$ or $t < 0$, where in each case it must b that φ' is strictly one sign (either ≥ 0 or ≤ 0), showing that φ *must* be an increasing function.

Initial Boundary Value Problem (IBVP)

Let us motivate this by going back to the transport equation but change up the restrictions:

$$\begin{cases} u_t + cu_x = 0 & x > 0,\ t > 0 \\ u|_{t=0} = f(x) & x > 0 \end{cases}$$

The general solution was found to be $u = \varphi(x - ct)$, and when plugging the initial data we get:

$$\varphi(x) = f(x)$$

as $x > 0$. Now, as f is only defined when $x > 0$, we have that u is defined when $x - ct > 0$. If c is negative (the wave propagates to the left) then we have no issue. However, if the wave propagates to the right, the wave shall run into a problem when $x < ct$. In this case, there is an extra condition we put to the *border* of our problem:

$$\begin{cases} u_t + cu_x = 0 & x > 0,\ t > 0 \\ u|_{t=0} = f(x) & x > 0 \\ u|_{x=0} = g(t) & t > 0 \end{cases}$$

This gives us that $u(0,t) = \varphi(-ct) = g(t)$ as $t > 0$, setting $x = -ct$ we get $\varphi(x) = g(\frac{-1}{c}x)$ when $x < 0$. Thus, we get that $u(x,t) = g(\frac{-1}{c}(x - ct)) = g(t - \frac{1}{c}x)$ as $x < ct$, giving a general solution of:

$$u = \begin{cases} f(x - ct) & x > ct \\ g(t - \frac{1}{c}x) & x < ct \end{cases}$$

Note that if $f(0) = g(0)$, we get a discontinuity (or no solution) since $x = ct$. There is still a way in which there is a valid solution, but it will have to wait for the notion of *Distribution*.

1.2 Multi-Dimensional

Let us now consider the same 1st order equations, but in the multidimensional case. These are essentially the same strategy, and hence some examples are just given to elucidate

Example 1.2.1: Simple Example

1. Consider

$$u_t + 2tu_x + xu_y = 0$$

This gives:

$$\frac{dt}{1} = \frac{dx}{2t} = \frac{dy}{x} = \frac{du}{0}$$

From the first equation, we get $x - t^2 = C$. Taking $dt/1 = dy/x$ and substituting $x = C - t^2$, we get:

$$y - Ct - \frac{t^3}{3} = D$$

Eliminating the constant C, we get:

$$D = y - (x - t^2)t - \frac{t^3}{3}$$

Having found the integral lines, we may finish off:

$$u(x,y,t) = \varphi(x - t^2, y - xt + \frac{2t^3}{3})$$

2. Consider

$$u_t + 2tu_x + xu_y = 12xtu$$

This gives:

$$\frac{dt}{1} = \frac{dx}{2t} = \frac{dy}{x} = \frac{du}{12xtu}$$

we get back the same:

$$x = t^2 + C \qquad y = Ct + \frac{t^3}{3}$$

Then:

$$\begin{aligned} \frac{du}{u} &= 12xtdt = (4t^3 + 12Ctdt) \\ \Rightarrow \ln(u) &= \ln(E) + (t^4 + 6Ct^2) \\ \Rightarrow u &= E\exp(t^4 + 6Ct) \\ \Rightarrow u &= E\exp(-5t^4 + 6xt^2) \end{aligned}$$

Hence, we get:

$$u = \exp(-6t^4 + 6xt^2)\varphi(x - t^2, y - xt + \frac{2t^3}{3})$$

For the multi-dimensional case, some more care must be given. (TBD)

2

wave equations

I found an excellent intuition on why the 2nd order derivative appears!

https://youtu.be/ly4S0oi3Yz8?t=428

I also this current opinion on how to think about $\frac{\partial^2}{\partial t^2} = \frac{\partial^2}{\partial x^2}$: Recall that $\frac{\partial^2}{\partial x^2}$ represents the average of the points around it (think of the 3b1b video on the heat equation). The $\frac{\partial^2}{\partial t^2}$ represents how points accelerate over time. Think of:

$$F = m\frac{d^2}{dt^2} = ma$$

where the a term represents how an object would accelerate (simply take $u(x,t)$ where $\frac{\partial^2}{\partial t} = c$ for some constant $c \in \mathbb{R}$ and see how it represents the "graph" uniformally accelerating (or moving at a constant speed if $\frac{\partial^2}{\partial t^2} = 0$)). Now, the fact that $\frac{\partial^2}{\partial t^2} = \frac{\partial^2}{\partial x^2}$ means that the acceleration of the point represents the average around any point, i.e. the concavity!

In terms of actually solving an wave equation, we will start by focusing on the case of:

$$0 = \frac{\partial^2}{\partial t^2} - \sum_i^n \frac{\partial^2}{\partial x_i^2} = \frac{\partial^2}{\partial t^2} - \nabla = \square$$

2.1 The d'Alembert's Formula

We'll do the 1d case first:

$$\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} = 0 \tag{2.1}$$

with initival value problem (IVP), also known as *Cauchy problem* of:

$$u(0,x) = f(x) \qquad \partial_t u(0,x) = g(x)$$

These two functions are the given data to find the existence and uniqueness of a solution to equation 2.1. What extra restriction that need to be added to f and g will be discussed later. It might seem obvious that some differentiation condition needs to be added, however after the answer is revealed, we'll see that we can get away with weaker assumptions.

With some clever manipulation, there is a solution $u(t, x)$ to this equation with respect to f and g. First, let

$$r := x + t \qquad s := x - t$$

Then changing the operators, we get:

$$2\partial_r = \partial_t - \partial_x \qquad -2\partial_s = \partial_t + \partial x$$

Thus, substituting this into equation 2.1, we get:

$$(\partial_t + \partial_x)(\partial_t - \partial_x)u = -4\partial_s\partial_r u = 0$$

which implies that $\partial_r u$ is independent of s, that is $\partial_r u = v(r)$. Thus, if $W(s)$ is our integration constant, integarting with respect to r, we get:

$$u(r, s) = \int^r v(r')dr' + W(s) := V(r) + W(s)$$

Thus, substituting back our original varaibles, we get:

$$u(t, x) = V(x + t) + W(x - t) \tag{2.2}$$

which can be thought of as decomposed u into a left-moving component and a right-moving component. We next want to express V an T in terms of our initial data. From equation 2.2, we can express the initial data at as:

$$\begin{aligned} u(0, x) = V(x) + W(x) = f(x) \\ \partial_t u(0, x) = V'(x) - W'(x) = g(x) \end{aligned}$$

Differentiating the first line with respect to x and combining the two equations to solve for V' and W', we get:

$$V'(x) = \frac{1}{2}f'(x) + \frac{1}{2}g(x) \qquad W'(x) = \frac{1}{2}f'(x) - \frac{1}{2}g(x)$$

Integrating with respect to x, we get:

$$\begin{aligned} V(x) = \frac{1}{2}f(x) + \frac{1}{2}\int_0^x g(x')dx' + c_1 \\ W(x) = \frac{1}{2}f(x) - \frac{1}{2}\int_0^x g(x')dx' + c_2 \end{aligned}$$

Since $c_1 + c_2 = 0$ so that we get equality with equation 2.2, we get that:

$$u(t, x) = V(x + t) + W(x - t) = \frac{1}{2}\left(f(x + t) + f(x - t)\right) + \frac{1}{2}\int_{x-t}^{x+t} g(x')dx' \tag{2.3}$$

This final form is known as *d'Alembert formula* for solutions of the one-dimensional wave equation. Notice that as we've defined equation 2.3, we only requite that $f, g \in C^0(\mathbb{R})$ (i.e. that they be continuous), or really $f, g \in L^1_{\text{loc}}$. This gives rise to a $u(t,x)$ that is *not* necessarily C^2. However, in order for $u(x,t)$ to be a solution, it must be twice differentiable, which would imply any slice would have to be twice differentiable, while $f, g \in L^1_{\text{loc}}$ would surely fail in general. All is not lost though, we will show later that such asolution is known as a *weak solution*, which we'll be getting to in chapter 3.

The next important remark we must make about this solution is that it only depends on some time-limited amount of data: the solution of $u(t,x)$ depends upon $(f(x'), g(x'))$ only over the interval $x - t \leq x' \leq x + t$ and on f at the points $x \pm t$. The set:

$$\{x + t = x'\} \cup \{x - t = x'\}$$

is called the *light cone*, and conversely given some x, the set

$$\{x' + t = x\} \cup \{x' - t = x\}$$

is called the *backwards light cone*.

Notice that given d'Alembert's formula, the solution $u(t,x)$ given above depnds on the initial data $(f(x), g(x))$ only for values x' that is inside the backwards light cone, or more precisely the interior of the light cone with boundary defined by the set above and the line $\{t = 0\}$:

PICTURE HERE

Thus, we say such a solution satisfies the principle of *finite propagation speed*. Thus, if f, g have compact support within some interval $[-R.R]$,

$$\operatorname{supp}(u(t,x)) \subseteq [-R - |t|, R + |t|]$$

as seen in figure BOOK IMAGE. if we consider the slope of the above to be c (notice that $c = 1$, then the information given by the initial data in the interval $[-R, R]$ does not travel faster then $c = 1$, and so at tiem $t > 0$, the information has not propogated further than $[-(R + t), (R + t)]$. This property is called *Huygen's principle*, and will appear often when we talk about solutions to wave equations.

(A word on the strong Huygen priniple)

2.2 Duhamel's Principle

We next look into the *inhomogeneous* 1d wave equation:

$$\partial_t^2 u - \partial_x^2 u = h(t,x) \tag{2.4}$$

with initial data

$$u(0,x) = f(x), \quad \partial_t u(0,x) = g(x), \quad h(t,x) \in C^2$$

If we have $h(t,x) = 0$, then we can find a solution $u_1(t,x)$ using d'Alembert's formula, and if $f = g = 0$, it is easy to get another [linear] solution $u_2(t,x)$, even if $h(t,x)$ is nonzero (as we'll explore momentarily). Thus, since equation (2.4) is a linear equation, we get a solution $u(t,x) = u_1(t,x) + u_2(t,x)$, gives the general solution! It remains to show how we can find u_2, which will be used using *Duhamel's principle*.

Start by solving the "standard form" of the IVP, namely with:

$$f = h = 0 \qquad \text{and} \qquad g = h$$

This comes from interpreting the h function as giving a new "force" on the surface (in this 1-dimensional case, string) which change the velocity, hence why $g = h$. Using d'Alembert's formula[1], we get:

$$W_t(g, x) = u(t, x) = \int_{x-t}^{x+t} g(s)ds$$

where we'll call $W_t(g, x)$, or just $W(t)$ the *propagation operator* or *solution operator*. Using this operator, we can show that the general solution is in fact:

$$u(t, x) = \partial_t(W_t(f, x)) = W_t(g, x)$$

We simply verify that this satisfies the initial conditions. For $u(0, x)$:

$$u(0, x) = \partial_t(W_0(f, x)) + W_0(g, x) = f(x) + 0$$

$$\partial_t u(0, x) = \partial_t\left(\partial_t(W_0(f, x))\right) + \partial_t(W_0(g, x)) = \nabla W_0(f, x) + g(x) = 0 + g(x)$$

Now that we expressed the d'Alembert's solution when the IVP is in standard form, we will use this to show how to solve the inhomogenous case. Letting $h \neq 0$, we'll show that:

$$u_2(t, x) = \int_0^t W(t-s)(h(s, \cdot))(x)ds$$

Indeed, verifying directly that $u_2(t, x)$ satisfies the IVP:

$$\begin{aligned}
u_2(t, x) &= 0 \\
\partial_t u_2(t, x) &= \partial_t \int_0^t W(t-s)(h(s, \cdot))(x)ds\bigg|_{t=0} = W(0)(h(t, \cdot))(x) = 0 \\
\partial_t^2 u_2(t, x) &= \int_0^t \partial_t^2 W(t-s)(h(s, \cdot))(x)ds + \partial_t\left(W(0)(h(t, \cdot))(x)\right) \\
&= \nabla u_2(t, x) + h(t, x)
\end{aligned}$$

Combining the result to get $u = u_1 + u_2$ and decomposing $W(t)(g)(x)$, we get:

$$\begin{aligned}
u(t, x) &= u_1(t, x) + u_2(t, x) \\
&= \frac{1}{2}\left(f(t+x) + f(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} g(x')dx' \\
&\quad \frac{1}{2}\int_0^t \int_{x-(t-s)}^{x+(t-s)} h(s, x')dx'ds
\end{aligned}$$

The double integral in the above takes into account the component $u_2(t, x)$ of the solutions over the backwards light cone

[1] you can veirfy through this link

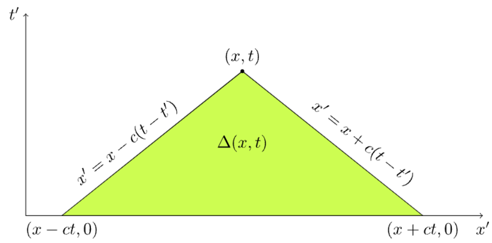

Figure 2.1: Light cone

2.2.1 Boundary Problems

Many times, a wave equation must also specify a boundary condition, thus we need to consider an *initial boundary problem*, or *IBP* for short. An example of a boundary problem would be :

$$u(t,0) = 0$$

which is known as the *Dirchelet condition*, and

$$\partial_x u(t,0) = 0$$

which is known as the *Neumann condition.* We will start by solving wave equations with the Dirchelet condition. We will take advantage of the fact that odd functions ($h(-x) = -h(x)$) that are continuous $x = 0$ must satisfy $h(0) = 0$. Let's assuem the sapcial domain is only $\mathbb{R}_{\geq 0}$, so that the inital values $f(x), g(x)$ are only definedon $\mathbb{R}_{\geq 0}$. Naturally, to be compatible with the IBP, we need that $f(0) = g(0) = 0$. We first define extension of f, g onto all of $\mathbb{R}$ by taking $f(-x) = -f(x)$, $g(-x) = -g(x)$. Using d'Alembert formula, we get the solution to the cauchy problem:

$$u(t,x) = \frac{1}{2}\left(f(x+t) - f(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} g(x')dx'$$

This expression is an odd fucntion with respect to x, since f and g are odd. Note that for any $x > t$, this expression only involves the arguments of f, g for positive x; however, for $x < t$, the expression involves negative argument even though $x > 0$. To remedy this we use the fact that he extended f, g are odd to rewrite the expression as:

$$u(t,x) = \frac{1}{2}\left(f(t+x) - f(t-x)\right) + \frac{1}{2}\int_{t-x}^{t+x} g(x')dx'$$

Thus, this solution is only dependnt upon $(f(x), g(x))$, $x > 0$. Due to the odd extension, the result looks like this acccross time:

(Then there is solving the Neumann condition, which since we are taking the derivative will end up being the same type of extension except it is an even extension instead of an odd one)

Example 2.2.1: Wave Equation Solutions

1. Consider:

$$\begin{cases} u_{tt} - 9u_{xx} = 0 & -\infty < t < \infty,\ x > -t \\ u|_{t=0} = 4\cos 9(x) & x > 0 \\ u_t|_{t=0} = -2\cos(3x) & x > 0 \\ u|_{x=-t} = u_x|_{x=-t} = 0 & 0 < t < \infty \end{cases}$$

Starting with the general solution, we get:

$$u(x,t) = \varphi(x+3t) + \psi(x-3t)$$

With the IVP, we get:

$$\begin{aligned} \varphi(x) + \psi(x) &= 4\cos(x) && x > 0 \\ \varphi(x) - \psi(x) &= -6\cos(x) && x > 0 \\ \Rightarrow \varphi(x) = -\cos(x) \qquad \psi(x) &= 5\cos(x) && x > 0 \end{aligned}$$

Hence the solution for this IVP with the right boundaries is:

$$u(x,t) = -\cos(x+3t) + 5\cos(x-3t) \qquad x > 3|t|$$

Now, for the boundary condition, we return to the original general solution and impose the boundary condition:

$$u(x,t) = \varphi(x+3t) + \psi(x-3t) \qquad u|_{x=-t} = u_x|_{x=-t} = 0 \qquad 0 < t < \infty$$

plugging in ,we get:

$$\varphi'(2t) = \psi'(-4t) = 0$$

For $t > 0$, we have φ defiend, so:

$$\sin(2t) + \psi'(4t) = 0 \Rightarrow \psi'(x) = \sin(x/2) \ \Rightarrow \ \psi(x) = -2\cos(x/2) + C$$

Hence for $t > 0$, $-t < x < 3t$, we get:

$$u(x,t) = -\cos(x+3t) - 2\cos((x-3t)/2) + 7$$

Doing the same for $t < 0$, where instead ψ is defined we get:

$$u(x,t) = \frac{5}{2}\cos(2x) - \frac{7}{2} + 5\cos(x-3t)$$

The solution boundaries looks like:

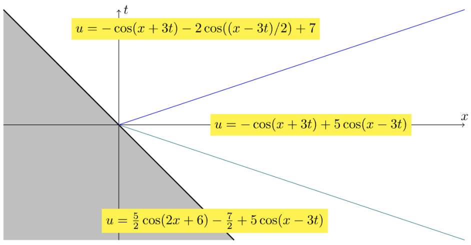

Figure 2.2: Where all the solutions are defined

This is also a nice visual: 3 reflection types.

2.2.2 Separation of Variables

Let:

$$\partial_t^2 - \partial_x^2 u = 0 \qquad x \in (0, 2\pi)$$

with Dirichlet boundary conditions:

$$u(t, 0) = 0 = u(t, 2\pi)$$

we will seek a special solution in the form of the product $u(t, x) = v(t)w(x)$ (i.e. the relation between the time and space variables can be separated to "independent" function which interact via multiplication) so that:

$$\partial_t^2 vw = v\partial_x^2 u$$

Assuming the measure of points where $v, w = 0$ is zero, dividing both sides by vw, we get that a.e.:

$$\frac{\partial_t^2 v}{v} = \frac{\partial_x^2 w}{w}$$

Since the left hand side only dependts on t while the right hand side only depends on t, the value of this equation is necessarily a constant:

$$\frac{\partial_t^2 v}{v} = \frac{\partial_x^2 w}{w} = \lambda$$

This makes the problem into 2 ODE's: an *evolution problem* for $v(t)$ in order to satisfy the initial conditions, and an eigenvalue problem for an ODE for $w(x)$ so that the solution satisfie sthe boundary condtiions. The eigenvalue probelm is for the pari $(w(x), \lambda)$, wher the function $w(x)$ satisfie the Dirichlet boundary conditions on $x \in (0, 2\pi)$:

$$\partial_x^2 w = \lambda w \qquad w(0) = 0 = w(2\pi)$$

Given an eigenfunctio nand an eiganvalue from this problem, the evolution equation for v is:

$$\frac{d^2}{dt^2} v = \lambda v$$

with initia ldat a $v(0) = a$, $\dot{v}(0) = b$ that has to be specified. The eigenvalue problem has a countable number of solutions, which are of the form:

$$w_k(x) = \frac{1}{\sqrt{\pi}} \sin\left(\frac{1}{2}kx\right) \qquad \lambda_k = \frac{-k^2}{4}, \ k \in \mathbb{N}_{>0}$$

Thus, we can choose countably many constants (a_k, b_k), $k \in \mathbb{N}_{>0}$ such that

$$v_k(t) = a_k \cos(w_k t) + b_k \frac{}{\sin(w_k t)} w_k$$

with temporal fequencies $w_k = \sqrt{-\lambda_k} = \frac{1}{2}k$. Combining these give us the general solution:

Theorem 2.2.2: General Solution To Seperation Of Variables

The genral solution of the initial-boundary value problem given at the beginning of this section is a lienar super position of these special solutiosn:

$$u(t,x) = \sum_{k=1}^{\infty} \left(a_k \cos(w_k t) + b_k \frac{\sin(w_k t)}{w_k} \right) \frac{1}{\sqrt{\pi}} \sin\left(\frac{1}{2}kx\right)$$

where the initail data is specified by:

$$u(0,x) = f(x) = \sum_{k=1}^{\infty} a_k \frac{1}{\sqrt{\pi}} \sin\left(\frac{1}{2}kx\right) = \sum_{k=1}^{\infty} a_k w_k(x)$$

and

$$\partial_t u(0,x) = g(x) = \sum_{k=1}^{\infty} b_k \frac{1}{\sqrt{\pi}} \sin\left(\frac{1}{2}kx\right) = \sum_{k=1}^{\infty} a_k w_k(x)$$

Furthermore, the sine series used to describe $f(x), g(x)$ is compelte as an orthonormal basis for the Hilbert space $L^2(0, 2\pi)$; thus any initial $f(x), g(x) \in L^2$ can be represented by their Fourier sine series cofefficeints $(a_k, b_k)_{k \in]N}$ giving rise to a solution $u(t,x)$.

***Proof*:**
Just take everything we talked about earlier. For the L^2 basis, recall 457 and 357.

To Calculate the coefficeints, we do the usual trick in Hilbert spaces given an orthonormal basis, recalling that the inner prodcut is:

$$\langle f, g \rangle = \int_0^{2\pi} f(x)\overline{g}(x)dx$$

2.3 Wave Equation in Higher Dimensions

we now study equations of the form:

$$\partial_t^2 u - \nabla u = 0$$

where $u \in \mathbb{R}^{1+n}$ and $\nabla u = \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2}$. This is often represented as:

$$\Box u = 0$$

2.3.1 Fourier Series

We first solve this using Fourier series. Assuming $f, g \in S$ is in the Schwartz class, we can take the Fourier Transform and get:

$$\partial_t^2 \hat{u}(t,\xi) + |\xi|^2 \hat{u}(t,\xi) = 0$$

making this into an ODE. Solutions of this ODE are composed of linear combinations of $e^{\pm i|\xi|t}$. Takiung into account that $\hat{u}(0,.\xi) = \hat{f}(\xi)$ and $\partial_t \hat{u}(0,\xi) = \hat{g}(\xi)$, we get the expression:

$$\hat{u}(t,\xi) = \cos(|\xi|t)\hat{f}(\xi) + \frac{\sin(|\xi|t)}{|\xi|}\hat{g}(\xi)$$

And so the inverse fourier transform gives the solution:

$$u(t,x) = \frac{1}{\sqrt{2\pi}^n}\int e^{i\xi x}\left(\cos(|\xi|t)\hat{f}(\xi) + \frac{\sin(|\xi|t)l}{|\xi|}\hat{g}(\xi)\right)d\xi$$

Thus, if f, g are of Schwartz class, we get a solution. However, this is by observation not necessary, we only need that:

$$\hat{f}(\xi), \frac{\hat{g}(\xi)}{|\xi|} \in L^2(\mathbb{R}^n_x)$$

to have a solution. We will soon explore this when we find weak solutions.

(a word on Lorenz Transfomration's leaving the solution independent, this is a prelude to chapter 6)

2.3.2 Method of Spherical Means

Definition 2.3.1: Spherical Mean

Given a function $g \in C(\mathbb{R}^n)$, it's *spherical mean* centered about the point $x \in \mathbb{R}^n$ is

$$M(h)(x,r) = \frac{1}{w_n r^{n-1}}\int_{|x-y|=r} h(y)dS_y$$

that is, it is the average of $h(x)$ over the sphere S^{n-1} centered at x of radius r.

the constant in front is the normalization term: $w_n r^{n-1}$ is the surface area of the sphere S^{n-1} of radius r of dimension n. When $n = 1$, the spherical mean of a function $f(x)$ is:

$$M(f)(x,r) = \frac{1}{2}(f(x+r) + f(x-r))$$

which is the first term the d'Alembert formula. If we do a change of variables in the equation in the definition to $y \mapsto x + r\xi$, where $\xi \in S_1(0)$, there is another expression for the spherical mean:

$$M(h)(x,r) = \frac{1}{w_n}\int_{|\xi|=1} h(x+r\xi)dS_\xi \tag{2.5}$$

Though we define if for $r \geq 0$, equation 2.5 shows that this function is even in r:

$$M(h)(x,-r) = M(h)(x,r)$$

Lemma 2.3.2: Darboux Equation

Given $h \in C^2(\mathbb{R}^n)$,

$$\triangle_x M(h)(x,r) = (\partial_r^2 + \frac{n-1}{r}\partial_R)M(h)(x,r)$$

a formula that realtes the Laplace operator of $M(h)$ in the n-dimensional x-variables to an operator involving only tone-dimeniwdimensional r derivatives.

***Proof*:**
This follows form computations using multivariable calculus. First, take one derivative:

$$\begin{aligned}\partial_r M(h)(x,r) &= \frac{1}{w_n} int_{|\xi|=1}\partial_r h(x+r\xi)dS_\xi \\ &= \frac{1}{w_n}\int_{|\xi|=1}\sum_{j=1}^n \partial_{x_j} h(x+r\xi)\xi_j dS_\xi\end{aligned}$$

Notice that $\sum_{j=1}^n \partial_{x_j} h(x+r\xi)\xi_j = \nabla h \cdot N$,the outwards noraml derivative of h on the unit sphere. Thus, we use Gren's theorem and continue:

$$\begin{aligned}\frac{1}{w_n}\int_{|\xi|=1}\sum_{j=1}^n \partial_{x_j} h(x+r\xi)\xi_j dS_\xi &= \frac{1}{w_n}\int_{|\xi|<1} r\triangle_x h(x+r\xi)d\xi \\ &= \frac{1}{w_n r^{n-1}}\triangle_x \int_{|x-y|<r} h(y)dy \\ &= \frac{1}{w_n r^{n-1}}\triangle_x \left(\int_0^r \int_{|x-y|=\rho} h(y)dS_y d\rho\right) \\ &= \frac{1}{r^{n-1}}\triangle_x \left(\int_0^r \rho^{n-1} M(h)(x,\rho)d\rho\right)\end{aligned}$$

the next second is to take a second derivagie after multiplying through by r^{n-1}:

$$\begin{aligned}\partial_r\left(r^{n-1}\partial_r M(h)(x,r)\right) &= \partial_r\left(\triangle_x \int_0^r \rho^{n-1}M(h)(x,\rho)d\rho\right) \\ &= \triangle_x\left(r^{n-1}M(h)(x,r)\right)\end{aligned}$$

Thus:

$$\begin{aligned}\triangle_x M(h)(x,r) &= \frac{1}{r^{n-1}}\partial_r\left(r^{n-1}\partial_r M(h)(x,r)\right) \\ &= \left(\partial_r^2 + \frac{n-1}{r}\partial_r\right)M(h)(x,r)\end{aligned}$$

as we sought to show.

A couple of analysis facts worth reminding ourselve:

Proposition 2.3.3: Analysis Facts

1. For $h(x) \in C(\mathbb{R}^n)$, the value of $h(x)$ at any $x \in \mathbb{R}^n$ can be recovered from tis spherical mean:
$$h(x) = \lim_{r \to 0} M(h)(x, r) = M(h)(x, 0)$$
For $h(x) \in C^2(\mathbb{R}^n)$,
$$\partial_r M(h)(x, 0) = \lim_{r \to 0} \frac{r}{w_n} \int_{|\xi|<1} h(x + r\xi) d\xi = 0$$

The first point will allow us to recoer the solution to wave equations posed in terms of the spherical mean ,while the other will help in computaion.

> ***Proof***:
> See [2].

We can now use the spherical means formula to find solutions to the wave equations, in particular solutions that are not as strong as being of Schwartz class. Next section, we will show how to find even weaker solutions. Firs,t suppose that $u(t, x)$ is a solution of the Cauch yproblem for the question equaiton. Then its spherical mean $M(u)(t, x, r)$ can be defiend from the equation for the spherical means, and it satisfies an auxiliary equation in the reduced space-time variables $(t, r) \in \mathbb{R}^2$. Specificaly, define:

$$M(u)(t, x, r) = \frac{1}{w_n} \int_{|\xi|=1} u(t, x + r\xi) dS_\xi$$

Taking time derivatves and using the fact that $u(t, x)$ is a solution of the wave equation, we obtain:

$$\begin{aligned} \partial_t^2 M(u)(t, x, r) &= \frac{1}{w_n} \int_{|\xi|=1} \partial_t^2 u(t, x + r\xi) dS_\xi \\ &= \frac{1}{w_n} \int_{|\xi|=1} \triangle_x u(t, x + r\xi) dS_\xi \\ &= \triangle_x M(u)(t, x, r) \end{aligned}$$

Now using the Darbous equation, we get:

$$\partial_t^2 M(u)(t, x, r) = \left(\partial_r^2 + \frac{n-1}{r} \partial_r \right) M(u)(t, x, r) \tag{2.6}$$

a PDE in the two variables $(t, r) \in \mathbb{R}^2$. This is known as the *Euler-Poisson-Darboux* equation.

2.3.3 Spherical Mean in 3 Dimension

We now show how to find solutiosn given the spherical Equation (2.6) is normally posed as an IVP:

$$\begin{aligned}
\partial_t^2 M(u)(t,x,r) &= \left(\partial_r^2 + \frac{n-1}{r}\partial_r\right) M(u)(t,x,r) \\
M(u)(0,x,r) &= M(f)(x,r) \\
\partial_t M(u)(0,x,r) &= M(g)(x,r)
\end{aligned}$$

The solution of this will vary quite a bit on the dimension we choose. The case of $n = 3$ is the simplest, and so we start with it. First, we do the substituion $M(u)(t,x,r) \mapsto rM(u)(t,x,r)$, giving:

$$\partial_t^2\left(rM(u)(t,x,r)\right) = r\left(\partial_r^2 + \frac{2}{r}\partial_r\right) M(u)(t,x,r) = \partial_r^2\left(rM(u)(t,x,r)\right)$$

Thus, the function $v(t,r) = rM(u)(t,x,r)$ is a solution of the wave equation in one dimension, and the variable x is relegated to the role of a parameter. The solution is ginve by d'Alembert formula

$$\begin{aligned}
v(t,r) &= rM(u)(t,x,r) \\
&= \frac{1}{2}\left((r+t)M(f)(x,r+t) + (r-t)M(f)(x,r-t)\right) \\
&\quad + \frac{1}{2}\int_{r-t}^{r+t} \rho M(g)(x,\rho)d\rho
\end{aligned}$$

Since $M(f)(x,r)$ and $M(g)(x,r)$ are even functions under $r \mapsto -r$, the n$rM(f)(x,r)$ and $rM(g)(x,r)$ are odd; therefore we may re-write the above expression, dividng through by r:

$$\begin{aligned}
M(u)(t,x,r) &= \frac{1}{2r}\left((r+t)M(f)(x,r+t) + (r-t)M(f)(x,r-t)\right) \\
&\quad + \frac{1}{2r}\int_{t-r}^{t+r} \rho M(g)(x,\rho)d\rho
\end{aligned}$$

Note that we used the fact that $\int_{t-r}^{r-t} \rho M(g)(x,\rho)d\rho - 0$, which holds since $rM(g)(x,r)$ is odd. Taking the limit as $r \to 0$ and using proposition 2.3.3, we see that we recover a representation for the solution $u(t,x)$.

Theorem 2.3.4: Kirchhoff's Formula

When $n = 3$, the solution to the wave equation is given by the expression

$$u(t,x) = \partial_t\left(tM(f)(x,t)\right) + tM(g)(x,t) \tag{2.7}$$

which is well defined as long as $f(x) \in C^1(\mathbb{R}^3)$ and $g(x) \in C(\mathbb{R}^3)$

***Proof*:**
Piece together all that was put above

Note that in terms of pointwise regularity, the solution is genreally less smooth than the initial data, for the Kirchhoff formula depends upon the derivativge of $f(x)$. In particular, carrying out the differentiation in equation (2.7), we obtain

$$u(t,x) = \frac{1}{4\pi t^2}\int_{|x-y|=t}\left(tg(y) = f(y) + t\nabla f(y)\cdot\frac{x-y}{|x-y|}\right)dS_y$$

Corollary 2.3.5: Solution Space

Givien initial data $g(x) \in C^2(\mathbb{R}^3)$ and $f(x) \in C^3(\mathbb{R}^3)$, the spherical means solutions given in equation (2.7) is a solution $u(t,x) \in C^2(\mathbb{R}^1_t \times \mathbb{R}^3_x)$

***Proof*:**
Immediate.

(comment on loss of differentiability, and how this doesn't happen in Sobolev space)

2.3.4 Spherical Means for odd Dimension

Let's first recall the Euler-Poisson Daboux equation for the n-dimensional wave equation with IVP:

$$\begin{aligned}
\partial_t^2 M(u)(t,x,r) &= \left(\partial_r^2 + \frac{n-1}{r}\partial_r\right) M(u)(t,x,r) \\
M(u)(0,x,r) &= M(f)(x,r) \\
\partial_t M(u)(0,x,r) &= M(g)(x,r)
\end{aligned}$$

The goal is to have an algebraic reduction of the wave equation to two dimensions in the varaibvles (t,r). This is carrid out in the odd-dimenisonal case as follows:

Lemma 2.3.6: Useful Relations

Suppose that $k \geq 1$ is an integer and that $h = h(r) \in C^{k+1}(R^1_+)$. Then:

$$\partial_r^2 \left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}h\right) = \left(\frac{1}{r}\partial_r\right)^k \left(r^{2k}\partial_r h\right)$$

$$\left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}h\right) = \sum_{j=0}^{k-1} \beta_j^k r^{j+1}\partial_r^j h$$

with the combinatorial coefficients being $\beta_0^k = 1 \cdot 3 \cdot \ldots \cdot (2k-1) := (2k-1)!!$.

***Proof*:**
Simply do an argument by induction

The ruslt is now useful since we can set $n = 2k+1$ and take:

$$v(t,x,r) := \left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}M(u)\right)(t,x,r)$$

For a give solution $u(t,x)$ of the wave equation. Then using the identies from corrollary 2.3.6

$$\begin{aligned}
\partial_r^2 v &= \partial_r^2 \left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}M(u)\right) \\
&= \frac{1}{r}\partial_r^k \left(r^{2k}\partial_r M(u)\right) \\
&= \left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}\partial_r^2 M(u) + 2kr^{2k-2}\partial_r M(u)\right) \\
&= \left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}\left(\partial_r^2 M(u) + \frac{2k}{r}\partial_r M(u)\right)\right) \\
&= \left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}\partial_t^2 M(u)\right) \\
&= \partial_t^2 v(t,r)
\end{aligned}$$

Thus, $v(t,x,r)$ is the quantity that satisfies the one-dimensional wave equation in the variables (t,r), for which we can applyh the d'Alembert formula:

$$v(t,x,r) = \frac{1}{2}\left(v(0,r+t) + v(0,r-t)\right) + \frac{1}{2}\int_{r-t}^{r+t} \partial_t v(0,x,\rho)d\rho$$

with initial data:

$$\begin{aligned}
v(0,x,r) &= \left(\frac{1}{4}\partial_r\right)^{k-1} \left(r^{2k-1}M(f)\right)(x,r) \\
\partial_t v(0,x,r) &= \left(\frac{1}{4}\partial_r\right)^{k-1} \left(r^{2k-1}M(g)\right)(x,r)
\end{aligned}$$

Then we can recover the solution as a limit:

$$u(t,x) = \lim_{r\to 0} M(u)(t,x,r) = \lim_{r\to 0} \frac{v(t,x,r)}{\beta_0^k r}$$

$$= \frac{1}{(n-2)!!}\left[\partial_t \left(\frac{1}{t}\partial_t\right)^{(n-3)/2} \left(t^{n-2}M(f)(t,x)\right) + \left(\frac{1}{t}\partial_t\right)^{(n-3)/2} \left(t^{n-2}M(g)(t,x)\right)\right]$$

Piecing it all together:

Theorem 2.3.7: Solution For Odd Dimension

For n odd, $f \in C^{(n-1)/2}(\mathbb{R}^n)$ and $g \in C^{(n-3/2}(\mathbb{R}^n)$, the solution formula given above gives a classical solution to the wave equation in $\mathbb{R}^n$

***Proof*:**
piece the above together

It is intereting that we can quantify the possibler loss of smoothness of the solution voer the initial data, made claer by the formula above the theorem. Indeed, the reduction process from $M(u)$ to v involves $k-1=(n-3)/2$ derivatives, and the limit involve sone derivative; therefore in general the solution will be less regular than the initial data by $k=(n-1)/2$ many derivatives.

2.3.5 Huygen's Principle

Just like in the one-dimenisonal case, the information of the wave equation only propagates finitely far. We will see that the strong Huygen principle will hold for odd dimensions but not for even. In the follwoing, we will show it for $n=3$:

Theorem 2.3.8: Huygen's Principle And Strong Huygen's Principle

Consider solutions to the wave equatio nfor $x \in \mathbb{R}^3$ and assume that the Cauchy data $f(x)$ and $g(x)$ are compactlysupported such that $\operatorname{supp}(f) \cup \operatorname{supp}(g) \subseteq B_R(0)$. Then:

1. **(Huygen's principle)**: The solution $u(t,x)$ has its support within the bounded region $B_{R+|t|}(0)$:
$$\operatorname{supp}(u(t,\cdot)) \subseteq B_{R+|t|}(0)$$
2. **(Strong Huygen's principle)**: Additionally, for $|t|>R$, for any space-time point (t,x) in the region inside the light cones given by $\{(t,x) \mid |x| \leq |t|-R\}$, again the solution vainshes; $u(t,x)=0$

***Proof*:**

1. The result follows from kirchhoff's Formula
2. When $|t|>R$, and $|x|<|t|-R$, again the backwards light cone emanating from the point (t,x) does not intersect the support of the initial dta, in this case because the sphere $\{y \mid |x=y|=|t|\}$ is too big and has passed outside of $B_R(0)$.

   This proof is geometric, only depending upon the fact that the solution is represented as a spherical mean. Therefore the result holds for solutions of the wave equation in arbitrary odd dimension, as shown in the equation above theorem 2.3.7.

(Something slightly stronger holds; see [1].)

2.3.6 Duhamel's Principle in Higher Dimensions

here

3

Weak Solutions and Distributions

We now deal with the case where solutions can be “weaker”. As we saw, d’Alembert formula works perfectly well if $f, g \in L^1_{\text{loc}}(\mathbb{R})$. However, the initial conditions *require* that f, g be differentiable (afterall, they are slices of the function $u(t, x)$ and $\partial_t u(t, x)$, which must be at least twice and once differentiable respectively). So is there some way to circumnavigate this? The content of the first section of this chapter covers precisely this. In the next section, we shall generalize the results and define a whole new object that will allow us to greatly broaden our notion of function that will allow us to differentiate function that are simply locally integrable!

Let’s start with the usual solution $u \in C^2$ on a time strip:

$$S_T = [0, T] \times \mathbb{R}^n$$

and assume that:

$$\Box u = F \qquad u|_{t=0} = f \qquad \partial_t u|_{t=0} = g$$

We are going to use the idea of a test function φ. If you want to think of it in a Folland way, $\varphi \in C_c^\infty(S_T)$, and we are defining a functional on $C_c^\infty(S_T)$ taking $\varphi \mapsto \int \Box u \varphi d(x, t)$. We will take advantage of integration by parts to get the following:

$$\begin{aligned}
\int_{S_T} F\varphi dt dx &= \int_{S_T} (\Box u)\varphi \, \mathrm{d}t\mathrm{d}x \\
&= \int_{\mathbb{R}^n} \left(-\int_0^T \partial_t u \partial_t \varphi \, \mathrm{d}t - g(x)\varphi(0, x) \right) \mathrm{d}x - \int_{S_T} u \triangle \varphi \, \mathrm{d}t\mathrm{d}x \\
&= \int_{S_T} u \Box \varphi \mathrm{d}t\mathrm{d}x + \int_{\mathbb{R}^n} f(x)\partial_t \varphi(0, x) \, \mathrm{d}x - \int_{\mathbb{R}^n} g(x)\varphi(0, x)\mathrm{d}x
\end{aligned}$$

Which leads us to the following definition:

Definition 3.0.1: Weak Solution

Let $f, g \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $F \in L_{\text{loc}}(S_T)$. We say $u \in L_{\text{loc}}(S_T)$ is a *weak solution* of the wave equation if:

$$\int_{S_T} (\Box u)\varphi \,\mathrm{d}t\mathrm{d}x = \int_{S_T} u\Box\varphi \mathrm{d}t\mathrm{d}x - \int_{\mathbb{R}^n} f(x)\partial_t\varphi(0,x)\,\mathrm{d}x + \int_{\mathbb{R}^n} g(x)\varphi(0,x)\mathrm{d}x \tag{3.1}$$

for all $\varphi \in C_c^\infty$ supported in $(-\infty, T) \times \mathbb{R}^n$

The next shows how this is indeed a generization:

Proposition 3.0.2: When Weak Solution Is Classical Solution

A weak solution that belongs to $C^2(S_T)$ is a classical solution

***Proof*:**
Taking equation (3.1), we see first that:

$$\int u\Box\varphi \, dtdx = \int F\varphi \, dtdx$$

and integratin by parts gives us that $\int u\Box\varphi = \int(\Box u)\varphi$. For the next part, recall that if $h \in L^1_{\text{loc}}$ and $\int h\varphi = 0$ for all $\varphi \in C_c^\infty$, then $h = 0$ almost everywhere. Thus since:

$$\int (\Box u - F)\varphi dtdx = 0$$

and φ was arbitrary, we conclude that $\Box u = F$ on S_T up to zero-measure.

Next, we need to show that u takes on the initial values of f and g. Let's first show that:

$$\partial_t u(0,x) = g(x)$$

First, the above equation is equivalent to:

$$\int_{\mathbb{R}^n} \partial_t u(0,x)\psi(x)dx = \int_{\mathbb{R}^n} g(x)\psi(x)dx \quad \text{for all } \psi \in C_c^\infty(\mathbb{R}^n) \tag{3.2}$$

Fix any such ψ. we will do some clever manipulation to "eliminate" the integral. Let $a(t)$ be the smooth step function going from 1 to zero smoothly:

$$a(t) = \begin{cases} 1 & \text{for } 0 \geq t \\ \text{smoooth} & \text{for } 0 \leq t \leq 1 \\ 0 & \text{for } t \geq 1 \end{cases}$$

Then $\theta_k(t) = a(kt)$ will squish the function so that it's closer and closer to being a step function. Importantly for us, notice that θ_k has the following properties:

$$\theta_k'(0) = 0 \qquad \theta_k(t) = 0 \quad \text{for } t \geq 1/k$$

Now, letting $\varphi = \varphi_k = \theta_k(t)\psi)$ from equation (3.1), then the right hand side of equation (3.1) reads as (taking k large enough so that $1/k < T$):

$$\int_{\mathbb{R}^n} \left(\int_0^{1/k} F(t,x)(akt)\, \mathrm{d}t \right) \psi(x) dx + \int_{\mathbb{R}^n} g(x)\psi(x) dx \tag{3.3}$$

Since $F \in C(S_T)$ (recall $\Box u = F$), the first term is $O(1/k)$ as $k \to \infty$. Now, I claim that the left hand side of equation (3.1) is equal to:

$$\int_{\mathbb{R}^n} \partial_t u(0,x)\psi(x)\, dx + O(1/k) \tag{3.4}$$

Notice that if (3.3) equals to (3.4), then as we take the limit as $k \to \infty$, we get equation (3.2). To show this is the case, notice the left hand side of (3.1) is:

$$\int_{S_T} u(t,x)\theta_k''(t)\psi(x) dt dx - \int_{\mathbb{R}^n} \left(\int_0^{1/k} u(t,x) a(kt) dt \right) \triangle psi(x) dx$$

The second term in the above is $O(1/k)$, and after integration by parts the first term becomes:

$$-\int_{S_T} \partial_t u(t,x)\theta_k'(t)\psi(x) dt dx$$

(Note that there are no boundary terms since $\theta_k'(t) = 0$ for $t = T = 0$). Doing integration by parts a second time, we get:

$$\int_{\mathbb{R}^n} \left(\int_0^{1/k} \partial_t^2 u(t,x) a(kt) dt \right) \psi(x) dx + \int_{\mathbb{R}^n} \partial_t u(0,x)\psi(x) dx$$

where again the first term is $O(1/k)$, which proves the claim, and thus completes the proof

(exercise doing the same for showing that $u(0,x) = f(x)$. It proceeds like the proof above, but this time choosing $\theta_k(t) = \frac{1}{k} b(kt)$ where b is some smooth compactly supported function such that $b(0) = 0$ and $b'(0) = 1$, thus $\theta_k(0) = 0$ and $\theta_k'(0) = 1$ and the support shrinks to the origin as $k \to \infty$ (see that the error term is not $O(1/k^2)$))

3.1 Distributions

Distributions are a generalization of functions that allow us to give more general solutions than classical solutions, similar to how we found a weak solution. Locally integrable functions will be distributions (in the appropriate interpretation), but we'll see that there are general solutions that don't even fall under the notion of "function".

The general theory of distributions, the space of test functions and its topology, the definition of a distribution as a continuous linear functional, the distributional derivative, and the basic operations on distributions, is developed in the functional-analysis volume, [3]; the Fourier theory of tempered distributions is developed in the harmonic-analysis volume, [4]. This chapter recalls the definitions it needs and concentrates on the role distributions play in the weak solutions of wave equations; it does not re-derive the foundational theory, which those volumes own.

We recall from [3] the notation and facts used below, none of them re-derived here. A *distribution* on an open $U \subseteq \mathbb{R}^n$ is a continuous linear functional on $C_c^\infty(U)$; their space is $\mathcal{D}'(U)$, with $\mathcal{D}' = \mathcal{D}'(\mathbb{R}^n)$, under the weak* topology. We write $\langle F, \varphi\rangle$ for the pairing and identify $f \in L^1_{\text{loc}}$ with $\varphi \mapsto \int f\varphi$; the point mass is $\langle \delta, \varphi\rangle = \varphi(0)$. Every $F \in \mathcal{D}'(U)$ has distributional derivatives $\langle \partial^\alpha F, \varphi\rangle = (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle$, may be multiplied by a smooth function and translated, and may be convolved with a test function ψ via $F * \psi(x) = \langle F, \tau_x\tilde{\psi}\rangle$ (so $\delta * \psi = \psi$). Finally $C_c^\infty(U)$ is dense in $\mathcal{D}'(U)$, and $\mathcal{D}'(U)$ is sequentially complete (a pointwise limit of distributions is again a distribution) by the uniform boundedness principle. Proofs and the continuity criterion are in [3, Distribution Continuity Check]; the inductive-limit topology on C_c^∞ itself is deliberately not constructed there (that book works with the seminorm test directly and defers the topology to its external references).

3.1.1 Time-dependent Distributions

Consider a map

$$\mathbb{R} \to \mathcal{D}'(\mathbb{R}^n), \qquad t \mapsto u_t$$

Since $\mathcal{D}'(\mathbb{R}^n)$ has a topology, the topology of point-wise convergence, the notion of continuity and differentiability can be defined. Thus, the above function is continuous at t_0 if and only if $t \to \langle u_t, \varphi\rangle$ is continuous for every $\varphi \in C_c^\infty(U)$, and u is differentiable at t_0 if and only if there exists a $v \in \mathcal{D}'(\mathbb{R}^n)$ such that for every $\varphi \in C_c^\infty(\mathbb{R}^n)$,

$$\frac{d}{dt}\,\langle u_t, \varphi\rangle|_{t=t_0} = \langle v, \varphi\rangle$$

and we write $u_t' = v$. This can naturally be repeated to form k-times differentiables maps in $C^k(\mathbb{R}, \mathcal{D}')$.

Proposition 3.1.1: Continuous Distributions Form Distributions

Every $u \in C(\mathbb{R}, \mathcal{D}')$ defines a distributions on $\mathbb{R}^{n+1}$ by

$$\langle u, \psi\rangle = \int_{\mathbb{R}} \langle u_t, \psi(t, \cdot)\rangle \qquad \text{for } \psi \in C_c^\infty(\mathbb{R}^n)$$

namely, $u \in \mathcal{D}'(\mathbb{R}^{1+n})$

***Proof*:**
First, the integrand is continuous and compactly supported as a function of t, and so the integral exists. The functional u is in $\mathcal{D}'(\mathbb{R}^{1+n})$ if and only if the all derivatives are properly bounded for all compact sets $K \subseteq \mathbb{R}^{1+n}$. Since the compact sets of the form $I \times K'$ are dense in the set of compact sets, it suffices to take $I \times K'$. But then, by the uniform boundedness principle on the Fréchet space $C_c^\infty(K')$ ([3, Principle Of Uniform Boundedness For Fréchet Spaces]), there exist $C_{I,K'} > 0$ and $N_{I,K'} \in \mathbb{N}$ such that

$$\sup_{t\in I} |\langle u(T), \varphi\rangle| \leq C_{I,K'} \sum_{|\alpha|\leq N_{I,K'}} \|\partial^\alpha \varphi\|_{L^\infty} \qquad \forall \varphi \in C_c^\infty(K')$$

Applying this to each $t \in I$ with $\varphi = \psi(t, \cdot)$, and then integrating with respect to t, we get the needed estimate to show it is bounded, showing it is indeed continuous, and so completing the proof.

With this build-up, we are now able to ask the following question: if $u, u' \in C^1(\mathbb{R}, \mathcal{D}')$, is it possible that:

$$\partial_t u = u'$$

with the derivative defined distributionally. The answer is yes! Unwarpping the definition, the above equation is equivalent to:

$$\int_{\mathbb{R}} \langle u(t), (-1)\partial_t \psi(t, \cdot)\rangle dt = \int_{\mathbb{R}} \langle u'(t), \psi(t, \cdot)\rangle dt$$

and since $u \in C^1$ (i.e. the derivative is continuous), then the derivative can be written in terms of distributions like so:

$$\frac{d}{dt}\langle u(t), \psi(t, \cdot)\rangle = \langle u'(t), \psi(t, \cdot)\rangle + \langle u(t), \partial_t \psi(t, \cdot)\rangle$$

To see this, note that if you take $f(t) = \langle u(t), \psi(t, \cdot)\rangle$, then

$$\begin{aligned} &\frac{1}{h}\{f(t+h) - f(t)\} \\ &= \left\langle \frac{1}{h}\left\{u(t+h) - u(t)\right\}, \psi(t, \cdot)\right\rangle \left\langle u(t+h), \frac{1}{h}\{\psi(t+h, \cdot) - \psi(t, \cdot)\}\right\rangle \end{aligned}$$

and then apply the uniform boundedness principle on the last term.

3.1.2 Distributional Solution for $\square u = 0$

We can now consider using distribution's to solve problems, namely:

$$\square u = 0 \qquad \Leftrightarrow \qquad \langle u, \square\varphi\rangle = 0 \qquad \forall \varphi \in C_c^\infty(U)$$

with the Cauchy problem:

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g$$

In general, it does not make sense to restrict a distribution u to the plane at $t = 0$ (what would this mean for a measure?). However, this *does* make sense for a time-dependent distribution as discussed in the last section. In fact, given *any* two distributions f, g, we can find a solution to the wave equation using time-dependent distributions!!

Theorem 3.1.2: General Weak Solution To The Wave Equation

For all $f, g \in \mathcal{D}'(\mathbb{R}^n)$, there exists a time-dependent distribution $u \in C^\infty(\mathbb{R}, \mathcal{D}'(\mathbb{R}^n))$ which solves the Cauchy problem

Note that we can replace $\mathbb{R}^n$ with U given we propertly separate U to be $U = I \times U'$. Furthermore, this solution is unique, which will be covered in the next section. The solution to u with respect to f and g is in fact extremely similar to D'Alembert's formula: we will have a *wave operator* $W(t)$ and will get the solution to be:

$$u(t) = W'(t) * f + W(t) * g \tag{3.5}$$

where

$$W \in C^\infty(\mathbb{R}, \mathcal{E}'(\mathbb{R}^n)) \qquad W(0) = W''(0) = 0 \qquad W'(0) = \delta$$

And furthermore, the Wave operator respects Huygen's principle:

$$\text{supp}W(t) \subseteq \{x \mid |x| \leq |t|\}$$

and if the dimension n is odd and $n \geq 3$:

$$\text{supp}W(t) \subseteq \{x \mid |x| = |t|\}$$

Before describing $W(t)$, let's show that an operator $W(t)$ satisfying the above condition does give a [weak] solution to the wave equation. First, since $W(t)$ is compactly supported, the convolution is well-defined for each t for all $f, g \in \mathcal{D}'(\mathbb{R}^n)$. Thus, equation (3.5) defines a smooth time-dependent distribution (check this). Next, the initial condition's are satisfied since:

$$u(0) = W'(0) * f + W(0) * g = \delta * f = f$$
$$u'(0) = W''(0) * f + W'(0) * g = \delta * g = g$$

Finally, to see that $\square u = 0$, we use an approximation argument. Since $C_c(\mathbb{R}^n)$ is dense in distributions, choose $f_i \to f$ and $g_i \to g$ in $\mathcal{D}'(U)$. Let u_i be the solution to the cauchy problem with f_i and h_i. Then:

$$u_i(t) = W'(t) * f_i + W(t) * g_i$$

Now, in $\mathcal{D}'(\mathbb{R}^{1+n})$:

$$u_i \to u$$

but then in $\mathcal{D}'(\mathbb{R}^{1+n})$:

$$\square u_i \to \square u$$

since $\square u_i = 0$ for all i, $\square u = 0$, completing the proof.

The verification of these details is left as an exercise.

We next tackle what the wave operator $W(t)$ actually is. (The d'Alembert and spherical-mean formulas realize this wave propagator; see [5].)

3.2 More on Distributional Solutions

We now move on to non-homogeneous solutions: we want to find a $u \in C^1([0,T], \mathbb{R}^n)$ that sovles

$$\begin{cases} \square u = F & \text{on} \qquad S_T = (0,T) \times \mathbb{R}^n \\ u|_{t=0} = f & \partial_t u|_{t=0} = g \end{cases}$$

Theorem 3.2.1: Unique Solution to Distributional Solution

The solution of the above equation is unique and of class $C^1([0,T], \mathcal{D}'(\mathbb{R}^n))$

***Proof*:**
See [5].

Proposition 3.2.2: Further Result about Weak Derivative I

Let $u, v \in C([0,T], \mathcal{D}'(\mathbb{R}^n))$ and set $S_T = (0,T) \times \mathbb{R}^n$. If

$$\partial_t u = v$$

i.e. u is the weak derivative of v, then $u \in C^1([0,T], \mathcal{D}'(\mathbb{R}^n))$ and $u' = v$

***Proof*:**
See [5].

Proposition 3.2.3: Further Results About Weak Derivatives II

Let $\Omega \subseteq \mathbb{R}^n$ be open. If $u \in C(\Omega)$ and the distributional derivative $\partial^\alpha u$ are also in $C(\Omega)$, for all $|\alpha| < k$, then $u \in C^k(\Omega)$

***Proof*:**
See [5].

Proposition 3.2.4: Consant Weak Derivative

Let $u \in \mathcal{D}'(I)$, where $I \subseteq \mathbb{R}$ is an open interval. Then:

$$u' = 0 \Rightarrow u \text{ is constant}$$

***Proof*:**
See [5].

3.2.1 A Fundamental Solution for the Square Operator

The forward fundamental solution E_+ of the wave operator $\Box$, characterized by $\Box E_+ = \delta$ and supported in the forward light cone, is constructed in [5].

3.2.2 Energy Identity

The energy identity gives a limits to how much the solution to a wave equation, or more generally a solution who is always on compact support which includes *inhomogeneous wave equations*, could have grown in a finite amount of time. In particular, we will bound $\|\nabla u(t, \cdot)\|_{L^2}$, by the "total" amount of potential energy that could have accumulated in the system. We first need to establish some notation and a lemma. Recall that the gradient of a differentiable function $u \in \mathbb{R}^{1+n}$ is:

$$\nabla u = (u_t, \nabla_x u)$$

which gives a basis for the tangent space of the codomain of u given the standard coordinates.

Lemma 3.2.5: Energy Identity in Homogeneous Case

Suppose $u \in C^2([0,T],\mathbb{R}^n)$ solves $\Box u = 0$, and that $u(t,\cdot)$ is compactly supported for every $t \in [0,T]$. Then:

$$\|\partial u(t,\cdot)\|_{L^2} = \|\partial u(0,\cdot)\|_{L^2}$$

***Proof*:**
Define the energy identity:

$$e(t) = \frac{1}{2}\int_{\mathbb{R}^n} |\partial u(t,x)|^2 \,\mathrm{d}x = \frac{1}{2}\|\partial u(t,\cdot)\|_{L^2}^2$$

Differentiating with respect to t and integrating by parts, we get:

$$e'(t) = \int_{\mathbb{R}^n} u_t \Box u dx$$

since $\Box u = 0$, we get that $e'(t) = 0$, and so $e(t) = c$ is a constant, and so $e(0) = e(t)$, which lets us conclude that:

$$\|\partial u(0,\cdot)\|_{L^2} = \|\partial u(t,\cdot)\|_{L^2}$$

as we sought to show.

We now move on to u which can be inhomogeneous. In fact, this proof works even more generally than solutions to wave equations, though as we will see it will naturally be applicable for wave equations

Theorem 3.2.6: Energy Inequality (Identity)

Let $u \in C^2([0,T],\mathbb{R}^n)$ and $u(t,\cdot)$ be compactly supported for every $t \in [0,T]$. Then:

$$\|\partial u(t,\cdot)\|_{L^2} \leq \|\partial u(0,\cdot)\|_{L^2} + \int_0^t \|\Box u(s,\cdot)\|_{L^2} ds$$

***Proof*:**
We start like how we've done in lemma 3.2.5: define the energy identity

$$e(t) = \frac{1}{2}\int_{\mathbb{R}^n} |\partial u(t,x)|^2 \,\mathrm{d}x = \frac{1}{2}\|\partial u(t,\cdot)\|_{L^2}^2 \tag{3.6}$$

Differentiating with respect to t and integrating by parts, we get:

$$e'(t) = \int_{\mathbb{R}^n} u_t \Box u \, dx \tag{3.7}$$

Now, apply the Cauchy-Schwarz inequality (equivalently Hölder's inequality) to equation (3.7), we get:

$$e'(t) \leq \|\partial u(t,\cdot)\|_{L^2}\|\Box u(t,\cdot)\|_{L^2} = \sqrt{2e(t)}\|\Box u(t,\cdot)\|_{L^2}$$

Thus, whenever $e(t) \neq 0$, we get:

$$\frac{d}{dt}\sqrt{e(t)} = \frac{e'(t)}{2\sqrt{e(t)}} \leq \frac{1}{\sqrt{2}}\|\Box u(t,\cdot)\|_{L^2}$$

Thus, integrating with $\int_0^t$, on both sides, we get:

$$\sqrt{e(t)} \leq \sqrt{e(0)} + \frac{1}{\sqrt{2}}\int_0^t \|\Box u(s,\cdot)\|_{L^2} ds$$

which, when substituting in what $e(t)$ equals to from equation (3.6), we get the inequality from the theorem, completing the proof.

3.2.3 Fourier Transform and Tempered Distributions

The Schwartz space, tempered distributions, and the Fourier transform on them are developed in the harmonic-analysis volume, [4]; we recall only what the wave-equation applications below require.

We now generalize earlier results of solving PDEs using Fourier transformations but use fourier transforms of distributions. Let $\mathcal{S}$ be the Schwartz space and $\mathcal{S}'$ is the dual of $\mathcal{S}$. Then naturally,

$$E' \subseteq \mathcal{S}' \subseteq \mathcal{D}'$$

where E' is the set of all compactly supported distributions. The dual $\mathcal{S}'$ is the space of *tempered distributions*, developed in [4, Tempered Distribution]; we use it freely below.

If $u \in \mathcal{S}'$, it has a *Fourier transform*:

$$\langle \hat{u}, \varphi \rangle = \langle u, \hat{\varphi} \rangle$$

If $u \in E'$, then $\hat{u}$ is a C^∞ function given by:

$$\hat{u}(\xi) = (2\pi)^{-n/2}\langle u, E_\xi \rangle$$

where $E_\xi \in C^\infty(\mathbb{R}^n)$ is $E_\xi(x) = e^{-ix\xi}$. To make sure that function multiplication still remains of Schwartz class, we will need a $\psi \in C^\infty(\mathbb{R}^n)$ to be slowly increasing, that is for all partial derivatives to have a growth at most:

$$|\partial^\alpha \psi(x)| \leq C_\alpha(1 + |x|)^{N(\alpha)} \qquad \forall \alpha$$

Then $\psi\varphi \in \mathcal{S}$ whenever $\varphi \in \mathcal{S}$. Thus, in this case, we can define for $u \in \mathcal{S}'$:

$$\langle \psi u, \varphi \rangle = \langle u, \psi\varphi \rangle$$

With this, we are ready to use Fourier transforms to solve the wave equation: let u be the solution of:

$$\Box u = 0 \quad \text{on } \mathbb{R}^{1+n}, \quad u|_{t=0} = f \quad \partial_t u|_{t=0} = g$$

with $f, g \in \mathcal{S}(\mathbb{R}^n)$. Applying the Fourier transform to the space of variables:

$$\hat{u}(t,\xi) = \frac{1}{(2\pi)^{n/2}}\int_{\mathbb{R}^n} e^{-ix\xi}u(t,x)dx$$

Then $\Box u = 0$ transforms into:

$$\partial_t^2 \hat{u}(t,\xi) + |\xi|^2 \hat{u}(t,\xi) = 0 \qquad \hat{u}(0,\xi) = \hat{f}(\xi) \qquad \partial_t \hat{u}(0,\xi) = \hat{g}(\xi)$$

But as before, this is just a second order ODE with respect to t with solution:

$$\hat{u}(t,\xi) = \cos(t|\xi|)\hat{f}(\xi) + \frac{\sin(t|\xi|)}{|\xi|}\hat{g}(\xi) \tag{3.8}$$

Now, we've seen earlier that:

$$u(t,\cdot) = W'(t) * f + W(t) * g$$

solve the wave equation. Since $W(t) \in E'$, we can apply the Fourier transform to the above equation and get:

$$\hat{u}(t,\xi) = (2\pi)^{n/2}\widehat{W'(t)}(\xi)\hat{f}(\xi) + (2\pi)^{n/2}\widehat{W(t)}(\xi)\hat{g}(\xi)$$

Comparing this with equation (3.8), we see that:

$$\widehat{W(t)}(\xi) = (2\pi)^{-n/2}\frac{\sin(t|\xi|)}{|\xi|} \qquad \widehat{W'(t)}(\xi) = (2\pi)^{-n/2}\cos(t|\xi|) \tag{3.9}$$

Giving us the full solution when $\Box u = 0$. For the inhomogeneous case, $\Box u = F$, with vanishing data at $t = 0$, we have:

$$u(t,x) = \int_0^t W(t-s) * F(s,x)ds$$

Applying the Fourier transform and applying equation (3.9), we get:

$$\hat{u}(t,\xi) = \int_0^t \frac{\sin((t-s)|\xi|)}{|\xi|}\hat{F}(s,\xi)ds$$

Thus, the full solution to the inhomogeneous case:

$$\Box u = F \quad \text{on } \mathbb{R}^{1+n}, \quad u|_{t=0} = f \quad \partial_t u|_{t=0} = g$$

is:

$$\hat{u}(t,\xi) = \cos(t|\xi|)\hat{f}(\xi) + \frac{\sin(t|\xi|)}{|\xi|}\hat{g}(\xi) + \int_0^t \frac{\sin((t-s)|\xi|)}{|\xi|}\hat{F}(s,\xi)ds$$

3.2.4 Sobolev Space H^s

So far, we found solutions to PDEs lie in $\mathcal{D}'$. This is great in finding general solutions, however the space $\mathcal{D}'$ doens't give much in the way of control of the regularity of functions (i.e. how much the derivative varries). In later studies of solutions to PDEs it will be quite important that we have some control over the derivative so that the solution "stays" nice. So, we can think of Sobolev spaces as spaces that

1. have sufficinetly many derivatives
2. the functions have sufficiently nice regularity properties.

The Bessel-potential construction of the $H^s(\mathbb{R}^n)$ scale (the operator $\Lambda^s = \mathcal{F}^{-1}(1+|\xi|^2)^{s/2}\mathcal{F}$, $H^s = \Lambda^{-s}L^2$ with $\|f\|_{H^s} = \|(1+|\xi|^2)^{s/2}\hat{f}\|_{L^2}$, the nesting $H^t \subseteq H^s$ for $s < t$, and the integer-order characterization $H^s = \left\{f \in L^2 \;\middle|\; \partial^\alpha f \in L^2,\ |\alpha| \le s\right\}$) is developed in [4, Sobolev Spaces (H^s), Properties of H^s]. We use these facts in the L^2 estimates below.

3.2.5 L^2 estimates for solutions of $\Box u = 0$

These L^2 estimates follow from the Fourier representation above together with Plancherel's theorem; see [5].

4

Non-Linear Equations

The goal of this chapter is to now generalize the conditions on which we find wave equations in *inhomogeneous* and *non-linear* space. We will first we will first allow f, g to be in H^s and H^{s-1} respectively instead of $\mathcal{D}'$, which wil guarantee enough regularity to find a solution in the *inhomogeneous* case. We will then generalize and try to find solution for the *non-linear* case, which we will show is in fact a much harder problem, and in contrast to the linear case there will not always exist a global solution.

The goal is to show that we can find *local solutions*. The space of functions that will be nice enough to find these solutions will turn out to be Sobolev spaces, and so they will play a prominent role in this chapter.

4.1 Existence and Uniqueness Inhomogeneous PDE

We first solve the Cauchy problem in the nicest of Sobolev spaces. Take the inhomogenous PDE:

$$\Box u = F \tag{4.1}$$

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g \tag{4.2}$$

Theorem 4.1.1: Unique Solution In Sobolev Space

Let $s \in \mathbb{R}$. Let $f \in H^s$ and $g \in H^{s-1}$ and $F \in L^1([0,T], H^{s-1})$. Then for every $T > 0$, there exists a unique solution u which belongs to

$$C([0,T], H^s) \cap C^1([0,1], H^{s-1}) \tag{4.3}$$

and solves equation (4.1) on $S_T = (0,T) \times \mathbb{R}^n$. Moreover, u satisfies:

$$\|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^{s-1}} \leq C_T \left(\|f\|_{H^s} + \|g\|_{H^s} + \int_0^1 \|F(t')\|_{H^{s-1}} dt' \right)$$

for all $0 \leq t \leq T$ where $C_T = C(1+T)$ and C only depends on s

***Proof*:**
This proof was broken down into the following exercises:

1. First, $F \in L^1([0,T], H^{s-1})$ means that

   $$[0,T] \to \mathbb{R} \qquad t \mapsto \|F(t)\|_{H^{s-1}}$$

   is in $L^1([0,T])$. Given such an F, it defines a distributionon S_T by:

   $$\langle F, \psi \rangle = \int_0^T \langle F(t), \psi(t, \cdot) \rangle dt \qquad \psi \in C_c^\infty(S_T)$$

   The key to see that the above integral converges, and defines an element of $\mathcal{D}'(S_T)$ is the following: we have the inequality:

   $$|\langle u, \varphi \rangle| \leq \|u\|_{H^s} \|u\|_{H^{-s}} \qquad \forall s \in \mathbb{R},\ u \in H^s,\ \varphi \in \mathcal{S}$$

   next, we have the inclusion:

   $$C_c^\infty \subseteq \mathcal{S} \subseteq H^s \subseteq \mathcal{S}' \subseteq \mathcal{D}'$$

   which are all sequentially continuous

2. Next, for any interval $I \subseteq \mathbb{R}$, we get the continuous inclusion:

   $$C(I, H^s) \subseteq C(I, \mathcal{S}') \subseteq C(I, \mathcal{D}')$$

   and similarly for every C^k. Thus, if u is in the solution space given in equation (4.3), then $u \in C^1([0,T], \mathcal{D}')$. This in fact proves uniqueness.

3. the remaining details are in [5]

For existence, (it is the standard approximation method, so for any $f, g \in \mathcal{S}$ and $F \in C^\infty([0,T], \mathcal{S})$, we have a solution. We will construct an appropraite Banach space and show that these will converge to the values in our solution space)

4.2 Non-linear Cauchy Problem

We now start looking at non-linear equations. We start by showing that it is *not* always the case that there is global existence!! Let

$$\square u = (\partial_t u)^2 \tag{4.4}$$

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} g \tag{4.5}$$

Then:

Theorem 4.2.1: Non-Global Solution to Wave Equation

For the above Cauchy problem, there exists $g \in C_c^\infty(\mathbb{R}^n)$ such that the above does *not* admit a C^2 solution past time T.

***Proof*:**
Take g to be a constant $c > 0$. Then equation (4.4) reduces to the ODE $u_{tt} = u_t^2$, or if we set $y = u_t$, we get:

$$y' = y^2 \qquad y(0) = c$$

Then the solution to this separable ODE IVP is:

$$y(t) = \frac{c}{1 - ct}$$

which blows up as $t \to 1/c$. Thus, $u(t, x) = -\log(1 - ct)$ sovles equation (4.4) with condition (4.5), but $u \to \infty$ as $t \to 1/c$

In order to know that this is indeed the only possible solution, we show uniqueness for solutions to equation (4.4):

Theorem 4.2.2: Uniqueness Of Above Wave Equation

Let $u \in C^2(\Omega)$ solve $\square u = (\partial_t u)^2$ in the solid backwards light cone

$$\Omega = \{(t, x) \mid 0 \leq t < T, |x - x_0| < T_t\}$$

with base $B_0 = \{x \mid |x - x_0| < T - t\}$, then u is uniquely determined by its data:

$$u|_{B_0} \qquad \partial_t u|_{B_0}$$

Thus, if u, v are two solutions to equation (4.4) with the same inital data in Ω, then $u = v$. To see this, notice that $u - v$ solves the *linear* equation

$$\square(u - v) = a(t, x)\partial_t(u - v) \qquad \text{in } \Omega$$

where $a = \partial_t u + \partial_t v \in C^1(\Omega)$, and that $u - v$ and $\partial_t(uv)$ vanish in B_0.

***Proof*:**
The decomposition above drives the proof, which is completed in the next section; see [5].

4.3 Local Existence and Uniqueness

We now study the general non-linear case. As we've seen, global solutions need not occur, however local existence solutions can occur. We will study exactly how these local existence solutions can happen in this section. Consider the Cauchy Problem:

$$\Box u = F(u, \partial u) \qquad F(0) = 0 \tag{4.6}$$

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g \tag{4.7}$$

In the following theorems, we will consider u and F as real-valued, but the results generalise easily to system valued functions (i.e. $\mathbb{R}^n$-valued), which we ommit for presentation clarity

We will present many theorems and show where they lead to: these present the core of what is needed for solving a non-linear PDE. However, these theorems are quite complex, and so we will state the narrative here, and in the next section we will solve them for a special but very important case. Then, in the next chapter, we will proceed with solving these theorems.

Theorem 4.3.1: Local Existence And Uniqueness

Let $s > \frac{n}{2} + 1$. Then for all $(f, g) \in H^s \times H^{s-1}$, there exists $T > 0$ and a unique

$$u \in C([0, T], H^s) \cap C^1([0, T], H^{s-1})$$

Solving the IVP in equation (4.6) on $S_T = (0, T) \times \mathbb{R}^n$. Furthermore, the time T can be chosen to depend continuously on $\|f\|_{H^s} + \|g\|_{H^{s-1}}$

We next might want to see when we can extend a solution to a time $T' > T$:

Theorem 4.3.2: Continuation Theorem

Let $s > \frac{n}{2} + 1$. Fix f, g as in theorem 4.3.1. Let $T^* = T^*(f, g)$ be the supremum of all $T > 0$ such that equation (4.6) has a solution on S_T satisfying equation (4.7). Then either $T^* = \infty$ or:

$$\partial u \in L^\infty(S_T)$$

Using these two, we can restrict the IVP functions to smooth compact functions and always find a solution:

Theorem 4.3.3: Smooth Solution

If $f, g \in C_c^\infty(\mathbb{R}^n)$. Then there exists a $T > 0$ and a unique

$$u \in C^\infty([0, T], \mathbb{R}^n)$$

solving equation (4.6) on S_T

By the uniqueness of a solution in the backwards light cone, we can generalize the previosu result to the following:

Corollary 4.3.4: Global Smooth Solution

Let $f, g \in C^\infty(\mathbb{R}^n)$. Then there exists a set

$$A = \{(t,x) \mid 0 \leq t \leq T(x)\}$$

where T is a continuous and stricitly positive function on $\mathbb{R}^n$, and a unique solution $u \in C^\infty(A)$ of equation (4.6).

***Proof*:**
See [5].

4.4 The model equation $\Box u = (\partial_t u)^2$

We now prove these theorems in the special case where:

$$\Box u = (\partial_t u)^2$$

We will need the following identities and inequalities:

1. *Energy Identity* presented in section 3.2.2
2. *Sobolev's Lemma* on $\mathbb{R}^n$, in particular the inequality:
$$\|f\|_{L^\infty} \leq C_s \|f\|_{H^s} \qquad \forall f \in H^s \quad s > \frac{n}{2}$$
3. The *calculus identity*
$$\|fg\|_{H^s} \leq C_s(\|f\|_{L^\infty}\|g\|_{L^\infty} + \|f\|_{H^s}\|\|g\|_{L^\infty})$$
for all $s \geq 0$ and all $f, g \in H^s \cap L^\infty$. This is a spcial case of *Moser's Identity* (which would be needed for the general case of solving a non-linear PDE)
4. A special case of *Gronwall's Lemma*: If $C_1, C_2 \geq 0$ are given constants, and A is a continuous non-negative function on $[0,T]$ such that
$$A(t) \leq C_1 + C_2 \int_0^t A(\tau) d\tau \qquad 0 \leq t \leq T$$
Then:
$$A(t) \leq C_1 e^{C_2 t} \qquad 0 \leq t \leq T$$
This theorem is easy to see if you recall in EYNTKA 457 that we have classified all translation invariant function with amplitude 1 in the section on fourier transforms.

We will prove these identities later in the book. One more identity we will use and prove now is Sobolev's lemma:

Theorem 4.4.1: Sobolev's Lemma

Let $s > k + \frac{n}{2}$ where k is a non-negative integer. Then:

$$H^s(\mathbb{R}^n) \subseteq C^k(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$$

where the inclusion is continuous. Even better:

$$\sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^\infty} \leq C_s \|f\|_{H^s}$$

where C_s is independent of f

***Proof*:**
First, consider the case when $k = 0$. The key is to use the Cauchy-Schwarz inequality: if $f \in H^s$ where $s > \frac{n}{2}$, then:

$$\int |\hat{f}(\xi)| d\xi = \int (1 + |\xi|^2)^{s/2} (1 + |\xi|^2)^{-s/2} |\hat{f}(\xi)| d\xi \leq C_s \|f\|_{H^s}$$

where

$$C_s^2 = \int (1 + |\xi|^2)^{-s} d\xi < \infty$$

Thus, by the Fourier inversion formula and Riemann-Lebesuge formula:

$$f \in C_0 \cap L^\infty$$

and furthermore:

$$\|f\|_\infty \leq \int |\hat{f}(\xi)| d\xi \leq C_s \|f\|_{H^s}$$

which establishes the case for $k = 0$. For $k > 0$, consider $f \in H^s$ where $s > k + \frac{n}{2}$. Then apply the special case we've just proved to:

$$\partial^\alpha f \in H^{s - |\alpha|}$$

so that $\partial^\alpha f \in C_0 \cap L^\infty$ and:

$$\|\partial^\alpha f\|_\infty \leq C_{s - |\alpha|} \|\partial^\alpha f\|_{H^{s - |\alpha|}} \leq C_s \|f\|_{H^s}$$

and so, from proposition 3.2.2 that $f \in C^k$.

As a consequence:

$$|f(x)| \leq C_s \|f\|_{H^s} \qquad s > \frac{n}{2}$$

which by the definition of the Sobolev norm means:

$$|f(x)| \leq C \sum_{|\alpha| \leq \frac{n+1}{1}} \|\partial^\alpha f\|_{L^2}$$

We now prove these theorems in the special case where:

$$\Box u = (\partial_t u)^2 \tag{4.8}$$
$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g$$

Theorem 4.4.2: Local Existence And Uniqueness for Model non-linear PDE

Let $s > \frac{n}{2} + 1$. Then for all $(f, g) \in H^s \times H^{s-1}$, there exists $T > 0$ and a unique

$$u \in C([0,T], H^s) \cap C^1([0,T], H^{s-1})$$

Solving the IVP in equation (4.8) on $S_T = (0,T) \times \mathbb{R}^n$. Furthermore, the time T can be chosen to depend continuously on $\|f\|_{H^s} + \|g\|_{H^{s-1}}$

***Proof*:**
Existence: The technique used in this proof is very important and will be used often. We will define a clever Cauchy sequence that will converge to our result. Set:

$$u_{-1} = 0$$

Then define $u_0, u_1, ...$ inductively by:

$$\Box u_i = (\partial_t u_{i-1})^2 \quad u_i|_{t=0} = f,\ \partial_t u_i|_{t=0} = g \tag{4.9}$$

Then for $T > 0$ define $X_T = C([0,T], H^s) \cap C^1([0,T], H^{s-1})$. Note that X_T is a Banach space with the norm:

$$\|u\|_{X_T} = \sup_{0 \le t \le T} (\|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^{s-1}})$$

Now, notice that:

$$x_{i-1} \in X_T \Rightarrow x_i \in X_T$$

Indeed, since $s - 1 > \frac{n}{2}$, Sobolev's lemma and the calculus inequality give us:

$$\|vw\|_{H^{s-1}} \le C_s \|v\|_{H^{s-1}} \|w\|_{H^{s-1}} \tag{4.10}$$

for all $v, w \in H^{s-1}(\mathbb{R}^n)$. Thus:

$$u_{i-1} \in X_T \Rightarrow (\partial_{u_{i-1}})^2 \in C([0,T], H^{s-1})$$

which by the local existence theorem in the best-regularity case ([5]), we get that each iteration in equation (4.9) has a unique solution in X_T. Thus, the sequence of iterates is well-defined in X_T for any $T > 0$. Our next goal is to show this sequence is Cauchy, provided a sufficiently small $T > 0$. The limit of this sequence will be the solution to equation (4.8) on $S_T = (0,T) \times \mathbb{R}^n$ since

$$\begin{aligned} u_i \to u \in X_T \quad &\Rightarrow \quad u_i \to u \quad \text{in } \mathcal{D}'(S_t) \\ &\Rightarrow \Box u_i \to \Box u \quad \text{in } \mathcal{D}'(S_T) \end{aligned}$$

and by equation (4.10):

$$\begin{aligned} u_i \to u \in X_T \quad &\Rightarrow \quad (\partial_t u_i)^2 \to (\partial_t u)^2 \quad \text{in } C([0,T], H^{s-1}) \\ &\Rightarrow \Box(\partial_t u_i)^2 \to (\partial_t u)^2 \quad \text{in } \mathcal{D}'(S_T) \end{aligned}$$

which shows the IVP is satisfied and so u will solve the Cauchy Problem.

We now proceed to show that this is indeed a cauchy sequence. We will break this down in two steps:

(**Step 1**) The sequence is bounded:

$$\|u_i\|_{X_T} \leq 2CE_s \qquad \text{for} \qquad i = 0, 1, 2, ... \tag{4.11}$$

provided that T is so small so that:

$$T \leq \frac{1}{8C^2 E_s}$$

where

$$E_s = \|f\|_{H^s} + \|g\|_{H^{s-1}}$$

and $C > 0$ is a constant which only depends on s and n. Let's say the bound in equation (4.11) holds for $j - 1$. Then if we apply the energy inequality (theorem 3.2.6) to the iterations (4.9) and use the estimates derivated in equation (4.10), we get:

$$\|u_i\|_{X_T} \leq Ce_s + CT\|u_{i-1}\|^2_{X_T} \leq Ce_s + CT(2CE_s)^2$$

where we get that C only depends on s and n if $T \leq 1$ (C comes from the energy inequlaity and the esimate form equation (4.10)). If then $T \leq \frac{1}{8C^2 E_s}$ holds, then the right hand side is bounde by $2CE_s$, and so we obtain the bound in equation (4.11) fo rall j by induciton. Since the case $j = -1$ holds trivially.

Step 2 We will show the sequence satisfies:

$$\|u_{i+1} - u_i\|_{X_T} \leq \frac{1}{2}\|u_i - u_{i-1}\|X_T$$

which will show it's Cauchy. To see this, note that:

$$\Box(u_{i+1} - u_i) = \partial_t(u_i - u_{i-1})(u_i + u_{i+1})$$

with vanishing initial data at $t = 0$. Thus, if we apply the energy inequality and use equation (4.10) and the uniform bound of equation (4.11) we get:

$$\begin{aligned} \|u_{i+1} - u_i\|_{X_T} &\leq CT\left(\|u_i\|_{X_T} + \|u_{i-1}\|_{X_T}\right)\|u_i - u_{i-1}\|_{X_T} \\ &\leq CT4CE_s\|u_i - u_{i-1}\|_{X_T} \end{aligned}$$

Thus, w eget the desried bound using the fact that $T \leq \frac{1}{8C^2 E_s}$, completing the proof for existence

Uniqueness Now suppose that $u, v \in X_T$ for soem $T > 0$ solve the Cauchy problem on $S_T = (0, T) \times \mathbb{R}^n$ with the identical data at $t = 0$. Then:

$$\Box(u - v) = \partial_t(u + v)\partial_t(u - v)$$

and thensetting

$$A(t) = \|(u - v)(t)\|_{H^{s-1}} + \|\partial_t(u - v(t))\|_{H^{s-1}}$$

we get by the energy inequality and the result form equation (4.10):

$$A(t) \le C \int_0^t A(t')\, dt' \qquad \text{for} \qquad 0 \le t \le T$$

for some constant C independent of t (note that C depends on the norms $\|u\|_{X_T}$ and $\|v\|_{X_T}$, but this is not a problem since u and v are considered fixed for this argument). Then by Gronwall's Lemma, $E(t) = 0$ for $0 \le t \le T$, and so $u = v$ in S_T, showing the solution is also unique and so completing the proof.

Theorem 4.4.3: Continuation Theorem for Model non-linear PDE

Let $s > \frac{n}{2} + 1$. Fix f, g as in theorem 4.3.1. Let $T^* = T^*(f,g)$ be the supremum of all $T > 0$ such that equation (4.6) has a solution on S_T satisfying equation (4.7). Then either $T^* = \infty$ or:

$$\partial u \in L^\infty(S_T)$$

***Proof*:**
Let $s > \frac{n}{2} + 1$, $f\ inH^s$, and $g \in H^{s-1}$. Then it is enough to prove that if $0 < T < \infty$ and

$$u \in C([0,T), H^s) \cap C^1([0,T), H^{s-1})$$

solves the model non-linear PDE opn $S_T = (0,T) \times \mathbb{R}^n$ and satisfies

$$\nabla u \in L^\infty(S_T)$$

then

$$\sup_{0 \le t < T} (\|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^s-1}) < \infty$$

Since then, by theorem 4.4.2, we can etend the solution to a time strip $[0, T+\epsilon] \times \mathbb{R}^n$, for some $\epsilon > 0$.

To prove this, define:

$$A(t) = \|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^{s-1}} \tag{4.12}$$

By the energy inequality and the calculus inequality, we get:

$$\begin{aligned} A(t) &\le C_{s,T}\left(\|f\|_{H^s} + \|g\|_{H^{s-1}} + \int_0^t \|\partial_t u(t')\|_{L^\infty} \|\partial_t u(t')\|_{H^{s-1}} dt' \right) \\ &\le C_{s,T}\left(\|f\|_{H^s} + \|g\|_{H^{s-1}} + \|\nabla u\|_{L^\infty(S_T)} \int_0^t \|A(t')dt' \right) \end{aligned}$$

and now using Gronwall's lemma gives equation (4.12), completing the proof.

Theorem 4.4.4: Smooth Solution for Model non-linear PDE

If $f, g \in C_c^\infty(\mathbb{R}^n)$. Then there exists a $T > 0$ and a unique

$$u \in C^\infty([0,T], \mathbb{R}^n)$$

solving equation (4.6) on S_T

***Proof*:**
Let $f, g \in C_c^\infty(\mathbb{R}^n)$. Then *a priori* we havethat $f, g \in H^s$ for every $s \in \mathbb{R}$. Fix $s_0 > \frac{n}{2} + 1$. Then by theorem 4.4.2, there exists a $T > 0$ and a unique solution:

$$u \in C([0,T], H^{s_0}) \cap C^1([0,T], H^{s_0-1})$$

Solving the general non-linear PDE problem on $S_T = (0,T) \times \mathbb{R}^n$. We also have from the same theorem that for every $s > s_0$ there exists a $T_s > 0$ such that

$$u \in C([0,T_s], H^s) \cap C^1([0,T_s], H^{s-1})$$

I claim that we can take $T_s = T$. Indeed, this follows from theorem 4.4.3 since by Sobolev's lemma and by the fact that u is in $C([0,T], H^{s_0}) \cap C^1([0,T], H^{s_0-1})$:

$$\nabla u \in L^\infty(S_T)$$

Thus

$$u \in C(0,T], H^{s_0}) \cap C^1([0,T], H^{s_0-1})$$

for every $s \geq s_0$. By Sobolev's lemma again, we get:

$$\partial_t^j \partial_x^\alpha u \in C([0,T], \mathbb{R}^n) \tag{4.13}$$

for $j \in \{0,1\}$ and all α. Since u solves the non-linear PDE, we have that:

$$\partial_t^2 u = \triangle u + (\partial_t u)^2 \tag{4.14}$$

Thus, using this and equation (4.13), we have:

$$\partial_t^2 \partial_x^\alpha u \in C([0,T], \mathbb{R}^n)$$

so equation (4.13) holds for $j \in \{0,1,2\}$ and all α. Applying ∂_t to both sides of equation (4.14), we get:

$$\partial_t^3 u = \triangle \partial_t u + 2\partial_t u \partial_t^2 u$$

and now from this we see that equation (4.13) holds for $j \in \{0,1,2,3\}$. If we keep taking successive time derivatives of the equation, we obtain equation (4.13) for all j by induciton. Thus, we have that:

$$u \in C^\infty([0,T], \mathbb{R}^n)$$

as we sought to show.

5

Littlewood-Paley Theory

As we know, L^2 is a very natural generaliation of a space with an inner product when working over spaces of functions that are measurable under a given measure. In this chapter, we will learn how we can generalize the results of L^2 spaces into L^p spaces. In particular, we will figure out how to decompose a function f into "localized" frequences (f_p), which we then can use results of L^2 theory.

5.1 Little-Paley Decomposition

the first thing we'll do is find a way to decompose an L^p function (really any function) so that each piece will be L^2 Let $\chi \in C_c^\infty(\mathbb{R}^n)$ such that

$$0 \leq \chi \leq 1 \qquad \chi(\xi) = \begin{cases} 1 & \text{if } |\xi| < 1 \\ 0 & \text{if } |\xi| \geq 1 \end{cases}$$

Then define $S_j : \mathcal{S}' \to \mathcal{S}$ by

$$\widehat{S_j f} = \chi(2^{-j}\xi)\hat{f}$$

for $j \in \mathbb{N}$, that is S_j takes only a certain ranges of frequencies. Equivalently, we can write

$$S_j = \mathcal{F}^{-1}\chi(2^{-j})\mathcal{F}$$

Lemma 5.1.1: decomposing Schwartz Functions

For every $\varphi \in \mathcal{S}$, $\chi(2^{-j})\varphi \to \varphi$ in $\mathcal{S}$ as $j \to \infty$

***Proof*:**
was left as eercise

Since $\mathcal{F}$ is an automorphism on $\mathcal{S}$, we get:

$$S_j f \to f \qquad \text{in } \mathcal{S}$$

for every $f \in \mathcal{S}$. this then implies that $S_j f \to f$ in $\mathcal{S}'$ for every $f \in \mathcal{S}'$. Next, we define the following decomposition similar to martingale-difference representation used in the proof of Azuma's inequality: define

$$\triangle_0 = S_0 \qquad \triangle_j = S_j - S_{j-1} \qquad j \in \mathbb{N}$$

Thus, we get $f = \sum_0^\infty \triangle_j f$ in $\mathcal{S}$ (resp. $\mathcal{S}'$) for every $f \in \mathcal{S}$ (resp. $f \in \mathcal{S}'$)

Definition 5.1.2: Littlewood-Paley Decomposition

The decomposition $\{\triangle_j f\}_0^\infty$ is called the *Littlewood-Paley decomposition* of f.

Notice that $\widehat{\triangle_j f} = \beta_j(\epsilon)\hat{f}$ where

$$\beta_0(\xi) = \chi(\xi) \qquad \beta_j(\xi) = \chi(2^{-j}\xi) - \chi(2^{1-j}\xi) \qquad j \in \mathbb{N} \tag{5.1}$$

Notice that by definition of χ, we get tha $0- \leq B_j \leq 1$ for all j, and so

$$\sum_0^\infty \beta_j^2 \leq \sum_0^\infty \beta_j = 1$$

Proposition 5.1.3: Littlewood-Paley Decomposition

Let $\{\triangle_j f\}_0^\infty$ be the littlewood-Paley decomposition of f. Then:

1. $\text{supp}\widehat{S_j f} \subseteq \{|\xi| \leq 2^j\}$
2. $\text{supp}\widehat{\triangle_j f} \subseteq \{2^{j-2} \leq |\xi| \leq 2^j\}$
3. $S_j f = 2^{jn}\psi(2^j \cdot) * f$ where $\hat{\psi} = \chi$
4. $\triangle_i f = 2^{jn}\varphi(2^j \cdot) * f$ where $\hat{\varphi} = \chi(\xi) - \chi(2\xi)$
5. $\|S_j f\|_{L^p} \leq C\|f\|_{L^p}$ where $C = \|\psi\|_{L^1}$
6. $\|\triangle_i f\|_{L^p} \leq C\|f\|_{L^p}$ where $C = \|\varphi\|_{L^1}$
7. $S_j f \in C^\infty$ if $f \in L^p$ for $1 \leq p \leq \infty$
8. $\triangle_j f \in C^\infty$ if $f \in L^p$ for $1 \leq p \leq \infty$

***Proof*:**

1. here

2. here
3. here
4. here
5. Follows (3) and Young's inequlaity

$$\|f * g\|_{L^p} \leq \|f\|_{L^1} \|g\|_{L^p} \qquad 1 \leq p \leq \infty$$

6. follows from (4) and Young's inequality
7. here
8. here

We will apply this theory to Sobolev spaces H^s. Recall that

$$H^s = \left\{ f \in \mathcal{S}' \mid \Lambda^s f \in L^2 \right\}$$

where $\Lambda^s = (I - \triangle)^{s/2}$. We shows that H^s is a Hilbert space with the inner product

$$\langle f, g \rangle_{H^s} = \langle \Lambda^s f, \Lambda^s g \rangle_{L^s} = \int (1 + |\xi|^2)^s \hat{f}(\xi) \overline{\hat{g}(\xi)} d\xi$$

giving us norm $\|f\|_{H^s} = \|\Lambda^s f\|_{L^2}$, and $\Lambda : H^s \to L^2$ is a unitary isomorphism. Being a hilbert space, a few easy observations can be done: if $\hat{f}$ and $\hat{g}$ are almost disjiont, then $f \perp g$ in H^s. H^s is self-dual via the map $g \mapsto \langle \cdot, g \rangle_{H^s}$. By the property of H^s, we can see that H^s is unitarily isomorphic to H^{-s} via the pairing $\langle \cdot, \cdot \langle : H^s \times \S \to \mathbb{C}$ through

$$|\langle f, g \rangle| = |\Lambda^s f, \Lambda^{-s} g| = \left| \Lambda^s f \Lambda^{-s} g \right| \leq \|f\|_{H^s} \|g\|_{H^{-s}}$$

for all $f \in H^s$ and $g \in \mathcal{S}$. Thus, the map $g \mapsto \langle \cdot, g \rangle$ extends to a linear map from H^{-s} into $(H^s)^*$, which is a surjective isometry in view of the self-duality of L^2.

With this overview, we now present some quick observations relating H^s and the theory developped at the beginning

1. $S_j f \to f$ in H^s for everfy $f \in H^S$, since:

$$\|S_j f - f\|_{H^s}^2 = \int [1 - \chi(2^{-j})]^2 (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi \to 0$$

by the DCT

2. If $2^{j-2} \leq |\xi| \leq 2^j$, then

$$(1 + |\xi|^2)^{s/2} \approx 2^{js}$$

in the senes that there is a constant $C_s > 0$, indepennet of j, such that

$$C_s^{-1} 2^{js} \leq (1 = |\xi|^2)^{s/2} \leq C_s 2^{js}$$

we then get that

$$\|\triangle_i f\|_{H^s} \approx 2^{js} \|\nabla_j f\|_{L^2}$$

3. In view of the above, we have
$$\triangle_j f \perp \triangle_k f \qquad \text{if} \qquad |j-k| \geq 2$$
relative to the inner product on H^s

With these, we have the following estimate:

Proposition 5.1.4: Estimating Sobolev Norm

Let $s \in \mathbb{R}$. Then for every $f \in \|H^s\|$,
$$\|f\|_{H^s}^2 \approx \sum_0^\infty 2^{2js} \|\triangle_j f\|_{L^2}^2$$

***Proof*:**
By the 2nd observation, we already have that:
$$\|f\|_{H^s}^2 \approx \sum_0^\infty \|\nabla_j f\|_{H^s}^2$$
Then, using orthogonality, our 3rd observation, and the Cauchy-Schwarz ineqluaity, we get
$$\begin{aligned}
\|f\|_{H^s}^2 = \langle f, f\rangle_{H^s} &= \left\langle \sum \Delta_j f, \sum \Delta_d f \right\rangle_{H^s} \\
&= \sum\sum \langle \Delta_j f, \Delta_k f\rangle_{H^s} \\
&= \sum_{l=-1}^{1}\sum_{j=0}^{\infty} \langle \Delta_j f, \Delta_{j+l} f\rangle_{H^s} \\
&\leq \sum_{l=-1}^{1}\sum_{j=0}^{\infty} \|\Delta_j f\|_{H^s}\|\Delta_{j+l} f\|_{H^s} \\
&\leq \sum_{l=-1}^{1} \left(\sum_{j=0}^{\infty} \|\Delta_j f\|_{Hs}^2\right)^{1/2} \left(\sum_{j=0}^{\infty} \|\Delta_{j+l} f\|_{H^s}^2\right)^{1/2} \\
&\leq 3\sum_{j=0}^{\infty} \|\Delta_j f\|_{H^s}^2
\end{aligned}$$
On the other hand, by the observations of what is the Fourier transform on $\Delta_j f$ in equation (5.1),
$$\begin{aligned}
\sum_0^\infty \|\delta_j f\|_{H^s}^2 &= \sum_0^\infty \int (1+|\xi|)^2 |\beta_j(\xi)\hat{f}(\xi)|^2 d\xi \\
&= \int \left[\sum_i^\infty \beta_j^2(\xi)\right] (1+|\xi|)^2 |\hat{f}(\xi)|^2 d\xi \\
&\leq \int (1+|\xi|)^2 |\hat{f}(\xi)|^2 d\xi \\
&= \|f\|_{H^2}^2
\end{aligned}$$

completing the proof.

Using this result, we can find what types of sequences in L^2 would converge in H^s. Let's say $\{f_i\}$ is a sequence in L^2 such that for soem $R > 0$,

$$\text{supp}\hat{f}_i \subseteq \{R^{-1}2^j \leq |\xi|R2^j\}$$

and $\sum 2^{2js}\|f)i\|_{L^2}^2 < \infty$. Then $f = \sum f_i$ converges in H^s, and

$$\|f\|_{H^s}^2 \approx \sum 2^{2js}\|f_i\|_{L^2}^2$$

If $s > 0$, we can get essentially the same result if we only assume

$$\text{supp}\hat{f}_i \subseteq \{|\xi| \leq R2^j\}$$

Proposition 5.1.5: Better Bounding Condition

Let $s > 0$. Suppose $f_i \in L^2$ satisfie s

$$\text{supp}\hat{f}_i \subseteq \{|\xi| \leq R2^j\}$$

for some $R \geq 1$, and that

$$\sum_0^\infty 2^{2js}\|f_j\|_{L^2}^2, \infty$$

Then $f = \sum f_j$ converges in H^s, and

$$\|f\|_{H^s}^2 \leq C_{s,R}\sum_0^\infty 2^{2js}\|f_j\|_{L^2}^2$$

For the proof of this proposition, we need a quick lemma

Lemma 5.1.6: build up lemma

Let $s < 0$ and $R \geq 1$. If $f \in H^s$ and

$$\text{supp}\hat{f} \subseteq \{|\xi| \leq R\}$$

then

$$\|f\|_{H^s} \leq 2^s R^s \|f\|_{L^2}$$

***Proof*:**
Simply notie that:

$$|\xi| \leq R \Rightarrow (1 + |\xi|^2)^s \leq (2R^2)^s$$

***Proof*:**
of proposition 5.1.5. See [5].

Note that if $R = 2^q$, then the constant $C_{s,R}$ is of the form $C_s 2^{2qs}$. This will be important for the proof of Mosers inequality.

5.2 Calculus Inequality

We know used what we've learned to prove the inequalities form section 4.4.

See [5].

5.3 Moser Inequality

Theorem 5.3.1: Moser Inequality

Assume that $F : \mathbb{R}^n \to \mathbb{R}$ is C^∞ and $F(0) = 0$. Then for all $s \geq 0$, there isa continuous function $\gamma = \gamma_s : \mathbb{R} \to \mathbb{R}$ such that

$$\|F(f))\|_{H^s} \leq \gamma(\|f\|_{L^\infty})\|f\|_{H^s}$$

for all $\mathbb{R}^n$-valued $f \in H^s \cap L^\infty$

this will have an intersting consequence:

Corollary 5.3.2: Moser Inequality Consequence

If $s > \frac{n}{2}$, the map $f \mapsto F(f)$, $H^s \to H^s$ is smooth

***Proof*:**
after we prove Moser inequality

To prove the Mosers, we will need a generalization of lemma 5.1.6. First, recall that the *spectrum* of f is the supprot of its Fourier transform.

Lemma 5.3.3: Bernstein's Lemma

Assuem that the spectrum of $f \in L^p$, $1 \leq p \leq \infty$, is conatined in the ball $|\xi| \leq 2^j$. Then:

$$\|\partial^\alpha f\|_{L^p} \leq C_\alpha 2^{j|\alpha|}\|f\|_{L^p}$$

for any multi-index α. Moreover, if the spectrum is contained in $2^{j-2} \leq |\xi| \leq 2^j$, then

$$C_k^{-1}2^{jk}\|f\|_{L^p} \leq \sup_{|\alpha|=k} \|\partial^\alpha f\|_{L^p} \leq C_k 2^{jk}\|f\|_{L^p}$$

for any $k \in \mathbb{N}$

***Proof*:**
this is essentially an application of Youngs inequlaity

***Proof*:**
of the Moser inequality. See [5].

5.4 Further Applications of Littlewood-Paley Theory

This is cool because it builds up to $H^{n/2}$ not being an algebra, and how it embeds into L^p or L^∞ under appropriate circumstances.

We first state without proving the following:

Theorem 5.4.1: Generalizing to L^p

If $2 \leq p < \infty$, then

$$\|f_{L^p}\|^2 \leq C \sum_0^\infty \|\triangle_i f\|_{L^p}^2$$

and if $1 < p \leq 2$, then:

$$\sum_0^\infty \|\triangle_i f\|_{L^p}^2 \leq C\|f\|_{L^p}^2$$

***Proof*:**
not proven.

With this, we will show the following:

Theorem 5.4.2: Embeddgins For H^s

If H^s is the a sobelev space, then:

$$H^s \hookrightarrow L^{\frac{2n}{n-2s}} \qquad 0 \leq s < \frac{n}{2} \tag{5.2}$$

$$H^s \hookrightarrow L^{\frac{2n}{n-2s}} s > \frac{n}{2} \tag{5.3}$$

***Proof*:**
FIrst, recall the poin mass δ belongs to H^{-s} when $s > n/s$. Thus:

$$|f(x)| = |\langle \delta(\cdot - x), f\rangle| \leq \|\delta(\cdot - x)\|_{H^s}\|f\|_{H^s} = \|\delta_{H^{-s}}\|f\|_{H^s}$$

proving equation (5.3). For equation (5.2), by Young's inequality we get:

$$\|\triangle_i f\|_{L^{\frac{2n}{n-2s}}} \leq C2^{js}\|f\|_{L^2}$$

and since $\triangle_i f = \triangle_i \left(\sum_{k=i-1}^{i+1} \delta_k f \right)$, we get:

$$\|\triangle_i f\|_{L^{\frac{2n}{n-2s}}} \leq C \sum_{k=i-1}^{i+1} 2^{ks} \|\triangle_k f\|_{L^2}$$

Now to complete the proof, squrae both sides, sum over j ,and use theorem 5.4.1 on the left hand side.

The embedding in equation (5.2) is precise

Example 5.4.3: Fails If $S \leq \frac{n}{2}$

We want ot show that $H^{n/2}$ is not a subset of L^∞, so assume $H^{n/2} \subseteq L^\infty >$ Then by the closed graph theorem, we have $H^{n/2} \hookrightarrow L^\infty <$ so

$$f\|_{L^\infty} \leq C\|f\|_{H^{n/2}}$$

but hteen, this implies that $\delta \in (H^{n/2})^* = H^{-n/2}$, which is false

Corollary 5.4.4: Not All Sobolev Spaces Form Algebras

$H^{n/2}$ is not an algebra

***Proof*:**

Assume that it is. Then $\|f^k\|_{H^{n/2}} \leq C^k \|f\|^k_{H^{n/2}}$ for all $K \in \mathbb{N}$. Thus, if $\|f\|_{H^{n/2}} < 1/C$, then $f^k \to 0$ in $H^{n/2}$ as $k \to \infty$. But this implies that some subsequence converges to zero a.e. on $\mathbb{R}^n$, so we must have $|f| < 1$ a.e. But this means that the ball of radius $1/C$ in $H^{n/2}$ is conatined in $L^\infty <$ whihc implies that H^s is a subset of L^∞, which contradicts example 5.4.3.

6

Global Existence for Non-Linear Wave Equation

We are now in a position to start undersatnding the general theorem for the global existence of a non-linear wave equation. It will turn out that being a dimension less than 4 makes it more complicated! As we've shown, a global solution doesn't always exist, however we can find an "almost global" solution by being able to scale the initial f and g by a chosen ϵ to extend the solution to as far as we'd like, with the extra condition that the non-linear term F starts out at the origin with no momentum.

The Klainerman-Sobolev inequality can be thought of what has to be "sacrificed" in order for the vector fields to commute. We can just simply commute them without changing the value of the norm, but we can bound how much it would change the value.

6.1 Goal

We are going to solve the following non-linear cauchy problem in full generality. In $\mathbb{R}^{1+n}$, let:

$$\Box u = F(\partial u) \tag{6.1}$$

$$u|_{t=0} = \epsilon f \qquad \partial_t u|_{t=0} = \epsilon g \tag{6.2}$$

where $F : \mathbb{R}^{1+n} \to \mathbb{R}$ is a given smooth functions which to second order at the origin:

$$F(0) = 0 \qquad DF(0) = 0$$

Our goal is to prove the following theorem:

Theorem 6.1.1: Global Existence Of Non-Linaer Wave Equation

Let $n \geq 4$. Let $f, g \in C_c^\infty(\mathbb{R}^n)$. Then there exists $\epsilon_0 > 0$ such that the two above equations has a solution

$$u \in C^\infty([0, \infty), \mathbb{R}^n)$$

provided $\epsilon \leq \epsilon_0$.

Note that ϵ_0 depends on f and g. Furthermore, the solution is unique due to the material in chapter 4.

For dimension $n \in \{1, 2, 3\}$, we will get (form the proof above) an asymptotic lower bounds on the lifespan:

$$T_\epsilon = T^*(\epsilon f, \epsilon g)$$

as $\epsilon \to 0$. Recall that the lifespan is the supremum of $T > 0$ such that the two equations at the begning has a solutoin $u \in C^\infty([0, T] \times \mathbb{R}^n)$. Specifically, we shall see that there exists a $c > 0$ st

$$\begin{aligned} T_\epsilon &\geq e^{c/\epsilon} && \text{if } n = 3 \\ T_\epsilon &\geq e^{c/\epsilon^2} && \text{if } n = 2 \\ T_\epsilon &\geq c/\epsilon && \text{if } n = 1 \end{aligned}$$

6.2 The Invariant Vector Fields

The Sovelev inequality states that:

$$|f(x)| \leq C_s \|f\|_{H^s} \qquad s > \frac{n}{2}$$

which by the definition of the Sobolev norm means:

$$|f(x)| \leq C \sum_{|\alpha| \leq \frac{n+1}{1}} \|\partial^\alpha f\|_{L^2}$$

This limit on the size of f with respect to to the growth of it's partial was important for the solution in section 4.4. We wil need a similar estimate on $u(t, x)$ in terms of the L^2 norm for derivatives of u multiplied by a decay factor of t.

Now, I'm not exactly sure what's going on here, but the space-time derivatives invovled are the following vector-fields:

$$\partial_t, \partial_1, ..., \partial_n \tag{6.3}$$

$$\Omega_{ij} = x_j \partial_i - x_i \partial_j \tag{6.4}$$

$$\Omega_{0j} = t\partial_i - x_i \partial_t \tag{6.5}$$

$$L_0 = t\partial_t \sum_1^n x_i \partial_i \tag{6.6}$$

Letting $x_0 := t$, we get consistent notation for all the vector-fields. The Ω_{ij} vector-fields have lot's of overlap by skew-symmetry, meaning we in fact have all these vector-fields if we only consider indexes $1 \leq i < j \leq n$. Thus, we have a total of :

$$(n+1) + \frac{n(n-1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2} + 1$$

different vector-fields. We can enumerate all these vecot-fields as :

$$\Gamma = (\Gamma_0, \Gamma_1, ..., \Gamma_m) \qquad m = \frac{(n+1)(n+2)}{2}$$

6.3 The Klainerman-Sobolev Inequality

The following will replace the Sobolev inequality:

Theorem 6.3.1: Klainerman-Sobolev Inequality

There is a constant C such that

$$(1+t+|x|)^{\frac{n+1}{2}} |u(t,x)| \leq C \sum_{|\alpha| \leq \frac{n+1}{2}} \|\Gamma^\alpha u(t,\cdot)\|_{L^2} \qquad \text{for} \qquad t > 0, x \in \mathbb{R}^n$$

whenever $u \in C^\infty([0,\infty) \times \mathbb{R}^n)$ and $\operatorname{supp}(u(t,\cdot))$ is compact for every t

***Proof*:**
left for page 67

Note that this theorem implies the result locally, that is for $0 < t < T$ with the same constant C if $u \in C^\infty([0,T) \times \mathbb{R}^n)$. To prove this reuslt, we will requires some lemmas. The first of these express that the *homogeneous* vector fields (6.4) (6.4) (6.4) span the tangent space of $\mathbb{R}^{1+n}$ outside the lightcone:

$$\Omega_{ij} = x_j \partial_i - x_i \partial_j$$
$$\Omega_{0j} = t\partial_i - x_i \partial_t$$
$$L_0 = t\partial_t \sum_1^n x_i \partial_i$$

Lemma 6.3.2: Klainerman-Sobolev Inequality Lemma I

For any multi-index $\alpha \neq 0$ and $(t,x) \notin \Lambda = \{(t,x) \mid |t| = |x|\}$ (i.e. not in the light-cone):

$$\partial^\alpha = \sum_{1 \leq |\beta| \leq |\alpha|} c_{\alpha\beta}(t,x)\Gamma^\beta$$

where $c_{\alpha\beta}$ are C^∞ and homogeneous of degree -$|\alpha|$ outside the ligthcone Λ. In fact, the sum of the right hand side only involves the homogeneous vector-fields.

***Proof*:**
See [5].

The following is a localized Sobolev Inequality

Lemma 6.3.3: Klainerman-Sobolev Inequality Lemma II

Given $\delta > 0$, there exists a constant C_δ such that

$$|f(0)|^2 \leq C_\delta \sum_{|\alpha| \leq \frac{n+2}{2}} \int_{|y|<\delta} |\partial^\alpha f(y)|^2 dy$$

for all $f \in C^\infty(\mathbb{R}^n)$. Moreover,

$$\sup_{0<\delta_0 \leq \delta} C_\delta < \infty$$

***Proof*:**
Fix a cut of $\chi \in C_c^\infty(\mathbb{R}^n)$ which is 1 in the unit ball at the origin. Applying

$$|f(x)| \leq C \sum_{|\alpha| \leq \frac{n+1}{1}} \|\partial^\alpha f\|_{L^2}$$

to the function

$$\chi(y/\delta) f(y)$$

yields the desired inequality with $C_\delta \leq C(1 + \delta^{-n-2})$. The final statemtn naturally follows.

Finally, the third lemma requires a Sobolev inequaltiy on the spherical means, that is we need a Sobolev inequality for a smooth function $v(q, \omega)$ where $q \in \mathbb{R}$ and $\omega \in S^{n-1}$. First, notice that the vectorfields Ω_{ij}, $1 \leq i < j \leq n$, can be seen as vector-fields on S^{n-1}. We can accordingly write for any point $\omega \in S^{n-1}$:

$$\partial_\omega^\alpha = \Omega_{1,2}^{\alpha_1} \cdots \Omega_{n-1,n}^{\alpha_m}$$

where $\alpha = (\alpha_1, \alpha_2, ..., \alpha_m)$ and $m = n(n-1)/2$

Lemma 6.3.4: Klainerman-Sobolev Inequality Lemma III

Given $\delta > 0$, there exists a constant C_δ such that

$$|v(q,\omega)|^2 \leq C_\delta \sum_{j+|\alpha| \leq \frac{n+2}{2}} \int_{|p|<\delta} \int_{\epsilon \in S^{n-1}} |\partial_q^j \partial_\omega^\alpha v(q+p, \epsilon) f(y)|^2 d\sigma(\eta) dp$$

for all $v \in C^\infty(\mathbb{R} \times S^{n-1})$. Moreover,

$$\sup_{0<\delta_0 \leq \delta} C_\delta < \infty$$

***Proof*:**
This follows from lemma 6.3.3 if we cover the sphere S^{n-1} by finitely many coordinate charts and choose a parition of unity. The key observation is that the vectorfields

$$\Omega_{1,2}, \Omega_{2,3}, ..., \Omega_{n-1,n}$$

spans $T_\omega S^{n-1}$ for all $\omega \in S^{n-1}$

With these, we can proceed with the proof of the Klainerman-Sobolev inequality:

***Proof*:**
of the Klainerman-Sobolev inequality If we have $t + |x| \leq 1$, then the inequality follows from the stanardd Sobolev inequality

$$|f(x)| \leq C \sum_{|\alpha| \leq \frac{n+1}{1}} \|\partial^\alpha f\|_{L^2}$$

and so we will assume $t + |x| > 1$. The argument will deped on whether (t, x) is lose to the lightcone or not.

(**Case 1**) Assume that:

$$|x| \leq \frac{t}{2} \qquad \text{or} \qquad |x| \geq \frac{3t}{2} \tag{6.7}$$

and define R to be:

$$R = t + |x| > 1$$

I claim that the coefficeints in lemmma 6.3.2 satisfy:

$$|y| \leq \frac{R}{8} \Rightarrow |c_{\alpha\beta}(t, x + y)| \leq C R^{-|\alpha|} \qquad \text{for} \qquad 1 \leq |\beta| \leq |\alpha| \leq \frac{n+2}{2}$$

Assuming this values holds, we can apply lemma 6.3.3 to the function $z \mapsto u(t, x + Rz)$ and get

$$\begin{aligned} |u(t,x)|^2 &\leq C \sum_{|\alpha| \leq \frac{n+2}{2}} \int_{|z| \leq \frac{1}{8}} \left| R^{|\alpha|} \partial_x^\alpha u(t, x + Rz) \right|^2 dz \\ &= C R^{-n} \sum_{|\alpha| \leq \frac{n+2}{2}} \int_{|z| \leq \frac{R}{8}} \left| R^{|\alpha|} \partial_x^\alpha u(t, x + Rz) \right|^2 dz \end{aligned}$$

where we did a change of varialbes to $y = Rz$. Now, using lemma 6.3.2 we have, for $\alpha \neq 0$,

$$\partial_x^\alpha u(t, x + y) = \sum_{1 \leq |\beta| \leq |\alpha|} c_{\alpha\beta}(t, x + y)(\Gamma^\beta u)(t, x + y)$$

we can get the result (using the assumption we made) that

$$R^n |u(t,x)|^2 \leq C \sum_{|\alpha| \leq \frac{n+2}{2}} \|\Gamma^\alpha u(t, \cdot)\|_{L^2}^2$$

showing that the inequality holds in the regions defined in equations (6.7) and by R. It now remains to show these are valid assumptions. We follows by using equation (6.7) and that

$$|y| \leq \frac{R}{8} \Rightarrow |t - |x+y|| \geq cR \quad \text{for some } c > 0$$

Rearranging, we get:

$$\left| \frac{t + |x+y|}{R} - 1 \right| \leq \frac{1}{8}$$

so the point $(t/R, (x+y)/R$ is in a compact set disjoint for mthe light cone Λ. Thus, our assumptions follow from the continuity and homogeneity of $c_{\alpha\beta}$

(**Case 2**) We know assume

$$\frac{t}{2} \leq |x| \leq \frac{3t}{2} \tag{6.8}$$

and $t + |x| > 1$. We will solve this by switching to polar coordiantes. Let

$$x = r\omega \qquad \text{where} \qquad r > 0,\ \omega \in S^{n-1}$$

Then we re-write the solutions as:

$$u(t,x) = u(t, r\omega) = v(t, r-t, w) = v(t, q, w)$$

where $q = r - t$, and $v(q, w)$ is a smooth function on S^{n-1}. Re-writing u in terms of v, we have that:

$$v(t,q,w) = u(t, (t+q)w)$$

This effects the derivative $\partial_q v$ which results in:

$$\partial_q v = \sum_1^n \omega_j \partial_j u = \partial_r u$$

furthermore, for $1 < i < j \leq n$, we have:

$$\Omega_{ij} u = \Omega_{ij} v$$

where on the left hand side we consider Ω_{ij} to act in ω as a vectorfield on S^{n-1}. Thus, with the restriction imposed in equation (6.8) ($\frac{t}{2} \leq |x| \leq \frac{3t}{2}$) and the third Klainerman-Sobolev lemma (lemma 6.3.4), when applied to $v(t, q, \omega)$, we get:

$$|u(t,w)|^2 = |v(t,q,\omega)|^2 \tag{6.9}$$

$$\leq C_t \sum_{j+|\alpha| \leq \frac{n+2}{2}} \int_{|p|<\frac{t}{4}} \int_{\eta \in S^{n-1}} |\partial_q^j \partial_\omega^\alpha v(t, q+p, \eta)|^2 d\sigma(\eta) dp \tag{6.10}$$

$$\overset{!}{\leq} C_t \sum_{j+|\alpha| \leq \frac{n+2}{2}} \int_{|p|<\frac{t}{4}} \int_{\eta \in S^{n-1}} \left|\partial_r^j \Gamma^\alpha u(t, (t+q+p)\epsilon)\right|^2 d\sigma(\eta) dp \tag{6.11}$$

where we used the fact that $\Omega_{ij}u = \Omega_{ij}v$ and $\partial_q v = \sum_1^n \omega_j \partial_j u = \partial_r u$ for the $\overset{!}{\leq}$ inequality. Next, notice we can in fact replace C_t by the C we found earlier, that is $C_t \leq C$, meaning was can find a constant C independent of t. Since:

$$|x| \leq \frac{3t}{2} \Rightarrow 1 < t + |x| \leq \frac{5t}{2} \Rightarrow t > \frac{2}{5}$$

we have that t is bounded away from 0, and so C_t is bounded above given the last statement of lemma 6.3.4.
Next, by the restrictin given for case 2, we have that $|q| = |r - t| \leq t/2<$ and $|p| \leq t/4$ implies that:

$$\frac{t}{4} \leq t + p + q \leq 2t$$

Thus, since $\partial_r u = \sum_1^n \eta_i \partial_i u$, and doing a chagne of variables to $r = t + p + q$ in equation 6.9, we get:

$$\begin{aligned}
|u(t,x)|^2 &\leq C \sum_{|\beta| \leq \frac{n+2}{2}} \sum_{\frac{t}{4} \leq r \leq 2t} \int_{\eta \in S^{n-1}} \left|\Gamma^\beta u(t, r\eta)\right|^2 d\sigma(\eta) dr \\
&\leq C t^{1-n} \sum_{|\beta| \leq \frac{n+2}{2}} \int_0^\infty \int_{\eta \in S^{n-1}} \left|\Gamma^\beta u(t, r\eta)\right|^2 d\sigma(\eta) r^{n-1} dr \\
&= C t^{1-n} \sum_{|\beta| \leq \frac{n+2}{2}} \left\|\Gamma^\beta u(t, \cdot)\right\|_{L^2}^2
\end{aligned}$$

Since $t \approx |x|$ and $t + |x| > 1<$ we have that this proves the Klainerman-Sobolev inequaltiy in the revious for case 2, completing the proof.

6.4 Proof of main theorem

We know go towards proving theorem 6.1.1, which as a reminder is:

Theorem 6.4.1: Global Existence Of Non-Linaer Wave Equation Reminder

Let $n \geq 4$. Let $f, g \in C_c^\infty(\mathbb{R}^n)$. Then there exists $\epsilon_0 > 0$ such that the two equations

$$\Box u = F(\partial u) \tag{6.12}$$

$$u|_{t=0} = \epsilon f \qquad \partial_t u|_{t=0} = \epsilon g \tag{6.13}$$

where $F : \mathbb{R}^{1+n} \to \mathbb{R}$ is a given smooth functions which to second order at the origin:

$$F(0) = 0 \qquad DF(0) = 0$$

has a solution

$$u \in C^\infty([0, \infty), \mathbb{R}^n)$$

provided $\epsilon \leq \epsilon_0$.

We will make a few observations wrapped into a lemma to make the proof easier:

Lemma 6.4.2: Build-Up For Existence Theorem

1. $|F(z)| \leq G_2(|z|)|z|^2$, $|DF(z)| \leq G_2(|z|)|z|$, $|D^m F(z)| \leq G_m(|z|)$ for $m \geq 2$ for all $z \in \mathbb{R}^{1+n}$, where $G_2, G_3, ...$ are continuous, increasing functions and $D^m F = \partial^\alpha F$ with $|\alpha| = m$
2. $\Box \Gamma^\alpha = \sum_{|\beta| \leq |\alpha|} c_{\alpha\beta} \Gamma^\beta \Box$ where $c_{\alpha\beta}$ are constants
3. For $\alpha \neq 0$, $\Gamma^\alpha[F(\partial u)]$ is a linear combination of terms:
$$[D^m F](\partial u) \Gamma^{\beta_1} \partial u \cdots \Gamma^{\beta_m} \partial u \tag{6.14}$$
where $1 \leq m \leq |\alpha|$ and $\sum_1^m |\beta_i| = |\alpha|$
4. In the above equation (equation 6.14), at most one β_i can have order $|\beta_i| > |\alpha|/2$. If we order the β_i so that $|\beta_m| = \max_{1 \leq i \leq m} |\beta_i|$, then
$$|\beta_i| \leq \frac{|\alpha|}{2} \qquad \text{for} \qquad 1 \leq i \leq m - 1$$
5. Let $N = n + 4$. If $|\alpha| \leq N$ and $|\beta_i| \leq |\alpha|/2$, then
$$|\beta_i| + 1 + \frac{n+2}{2} \leq N$$

***Proof*:**

1. Since F vanishes in the first order, we get lots of information. Set
$$G_m(r) = \sup_{|z| \leq r} |D^m F(z)|$$

Then G_m is continuous and increasing, and we get the 3rd equation holds. Now write

$$\partial_j F(z) = \partial_j f(z) - \partial_j F(0) = \int_0^1 \frac{d}{dt}\left[\partial_j F(tz)\right] dt = \left[\int_0^1 \nabla \partial_j F(tz) dt\right] \cdot z$$

if we take the absolute value, we get the second equation. Finally, applying the same argumetn to $F(z)$ gives:

$$|F(z)| \leq \sup_{0\leq t\leq 1} |DF(tz)||z| \leq G_2(|z)|z|^2$$

giving us the first equation, completing the proof.

2. This follows from the conditions we have on these vector fields, namely:

$$[\Box, \partial_i] = 0 \qquad 0 \leq i \leq n$$

$$[\Box, \Omega_{ij}] = 0 \qquad 0 \leq i < j \leq n$$

$$[\Box, L)0] = 2\Box$$

3. this is a matter of induction
4. this result is immedaite
5. $N/2 + 1 + (n+2/2) \leq N$ if and only if $N + 4 \leq N$

With all of this done, we are ready to tackle the main theorem

***Proof*:**
of the main theorem: We startwith some reductions. Set $N = n + 4>$ Define

$$A(t) = \sum_{|\alpha|\leq N} \|\Gamma^\alpha u(t,\cdot)\|_{L^2} \qquad 0 \leq t < T$$

whenever $u \in C^\infty([0,T)\times\mathbb{R}^n)$ solves the equations in theorem 6.4.1 on $[0,T)\times\mathbb{R}^n$ for $T > 0$. By the seonc dequation ,we get

$$A(0) \leq \frac{A\epsilon}{2} \tag{6.15}$$

where A depends only on f and g and their derivatives. I lcaim that there exists a $\epsilon_0 > 0$ such that if $T > 0$ and $C^\infty([0,T)\times\mathbb{R}^n)$ sovles the equations on $[0,T)\times\mathbb{R}^n$ with $\epsilon \leq \epsilon_0<$ then $A(t) \leq A\epsilon$ for all $0 \leq t < T$. To see this, observe that by the Sovelev inequality:

$$\|\partial u\|_{L^\infty([0,T)\times\mathbb{R}^n)} \leq C \sup_{0\leq t<T} A(t)$$

Therefore, if the claim holds, it follows (see [5]) that the lifespan $T_\epsilon = \infty$ whenever $\epsilon \leq \epsilon_0$.

Our next reduction is to prove the claim on the set

$$E = \{t \in [0,T) \mid A(s) \leq A\epsilon \text{ for all } 0 \leq s \leq t\}$$

Since $A(0) \leq A\epsilon/2$, E is nonempty. Since $A(t)$ is continuous with respect to t, E is relatively closed in $[0,T)$ (i.e. closed in the subspace topology induced by $[0,T)$). If we can show that E

Is relatively open (open in the subspace topology of $[0, T)$), then it will follow that $E = [0, T)$, and so we proceed with proving E is relatively open.

Fix $t_0 \in E$ wit h$T_0 < T$> Since $A(t)$ is continuous, there eixtss a $t_1 > t_0$ such that

$$A(t) \leq A\epsilon \qquad \text{for} \qquad 0 \leq t \leq t_1$$

This is known as the *boot-strap assumption.* We will prove this implies

$$A(t) \leq A\epsilon \qquad \text{for} \qquad 0 \leq t \leq t_0$$

for ϵ sufficiently small. IT suffices to prove that

$$A(t) \leq \frac{A\epsilon}{2} + C_A \epsilon \int_0^t \frac{A(s)}{(1+s)^{\frac{n-1}{2}}} ds$$

Then by GRonwall's Lemma we have:

$$A(t) \leq \frac{A\epsilon}{2} \exp\left[C_A \epsilon \int_0^t \frac{ds}{(1+s)^{\frac{n-1}{2}}} \right]$$

Sincethe integral in the above equation is finite when $n \geq 4<$ we only needto choose $\epsilon > 0$ so so small that

$$\exp\left[C_A \epsilon \int_0^t \frac{ds}{(1+s)^{\frac{n-1}{2}}} \right] \leq 2$$

which completes the proof

With these simplification, we are ready to prove the main result. Given that

$$[\Gamma_i, \partial_j] = \sum_{k=0}^{n} a_{ijk} \partial_k$$

we can apply the energy inequatliy, obtainign

$$\begin{aligned}
A(t) &\leq A(0) + C_N \int_0^t \sum_{|\alpha| \leq N} \|\Box \Gamma^\alpha u(s, \cdot)\|_{L^2} ds \\
&\leq \frac{A\epsilon}{2} + C_N \int_0^t \sum_{|\alpha| \leq N} \|\Gamma^\alpha \Box u(s, \cdot)\|_{L^2} ds \\
&\overset{!}{=} \frac{A\epsilon}{2} + C_N \int_0^t \sum_{|\alpha| \leq N} \|\Gamma^\alpha [F(\partial u)](s, \cdot)\|_{L^2} ds
\end{aligned}$$

where $\overset{!}{=}$ comes from the fact that $A(0) \leq \frac{A\epsilon}{2}$ and lemma 6.4.2(2), and C_N denotes a generic constant which can change form line to line. Next ,we estimate the value inside the integral and summand, that is:

$$\|\Gamma^\alpha [F(\partial u)](s, \cdot)\|_{L^2} \qquad |\alpha| \leq N$$

If $\alpha = 0$, we use the fact that $|F(z)| \leq G_2(|z|)|z|^2$ to get

$$\|F(\partial u)(t,\cdot)\|_{L^2} \leq G_2(\|\partial u(t,\cdot)\|_{L^\infty})\|\partial u(t,\cdot)\|_{L^\infty}\|\partial u(t,\cdot)\|_{L^2} \tag{6.16}$$

The first facto on the right hand side is bounded by a continuous function of A since by Sobolev's inequality and the fact that $A(t) \leq 2A\epsilon$ (where we can assume $\epsilon \leq 1$):

$$\|\partial u(t,\cdot)\|_{L^\infty} \leq CA(t) \leq 2CA\epsilon \tag{6.17}$$

By the Klainerman-Sobolev inequality, the second factor in equation (6.16) is bounded by

$$C\frac{A(t)}{(1+t)^{(n-1)/2}}$$

and by the fact that $A(t)|le2A\epsilon$, the third factor is bounded by $2A\epsilon$.

If $\alpha \neq 0$, then by lemma 6.4.2(3) we can rewrite $\Gamma^\alpha u(t,\cdot)$ as:

$$[D^m F](\partial u)\Gamma^{\beta_1}\partial u \cdots \Gamma^{\beta_m}\partial u$$

where we bound the L^2 norms in the space by

$$\|[D^m F]\,(\partial u(t,\cdot))\|_{L^\infty}\prod_{i=1}^{m-1}\|\Gamma^{\beta_i}\partial u(t,\cdot)\|_{L^\infty}\|\Gamma^{\beta_m}\partial u(t,\cdot)\|_{L^2} \tag{6.18}$$

We will now get our result when we consider the case of $m \geq 2$ and $m = 1$. If $m \geq 2$, then since $|D^m F(z)| \leq G_m(|z|)$ equation (6.17) we just proved, the first factor is bounded by a continuous function A. Then by the fact that $A(t) \leq 2A\epsilon$ and the fact that $|\beta_m| \leq |\alpha| \leq N$, we see that the last factor is bounded by $2A\epsilon$. Since we may assume that lemma 6.4.2(4) and (5) holds, we have that $|\beta_i| + 1|(n+2)/2 \leq N$ for $1 \leq i \leq m-1$. Hence, by the Klainerman=Sobolev inequality and $A(t) \leq 2A\epsilon$,

$$\prod_{i=1}^{m-1}\|\Gamma^{\beta_i}\partial u(t,\cdot)\|_{L^\infty} \leq C\left[\frac{A(t)}{(1+t)^{\frac{n-1}{2}}}\right]^{m-1} \leq CA^{m-2}\frac{A(t)}{(1+t)^{\frac{n-1}{2}}}$$

and so, we conclude that equation (6.18) is bounded by

$$C_A\epsilon A(t)(1+t)^{-(n-1)/2}$$

whne $m \geq 2$. When $m = 1<$ we get the same bound if we use $|DF(z)| \leq G_2(|z|)|z|$ insead, completing the proof.

6.5 Looking at Low Dimensions Solutions

We have limited ourselves to the case where $n \geq 4$. The case for lower dimensions is harder, and we start addressing it in this section.

First of, if $n \in \{1, 2, 3\}$, then the global existence fails in general (we'll show some later). Fortunately, the proof used for the $n \geq 4$ case gives asymptotic lower bounds on the lifespan: if T is the supremum of all lifespance, we have:

$$T_\epsilon = T^*(\epsilon f, \epsilon g)$$

as $\epsilon \to 0$. Specifically, we'll see that there exists a $c > 0$ such that

$$T_\epsilon \geq \begin{cases} e^{c/\epsilon} & n = 3 \\ c/\epsilon^2 & n = 2 \\ c/\epsilon & n = 1 \end{cases}$$

for sufficinetly small ϵ, with c depending on f and g.

6.5.1 Proof of lower bounds

For some $T > 0$, Let $u \in C^\infty([0,T) \times \mathbb{R}^n)$ solve

$$\Box u = F(\partial u)$$

$$u|_{t=0} = \epsilon f, \quad \partial_t u|_{t=0} = \epsilon g$$

where $F(0) = DF(0) = 0$. Recall from the proof of global existence for $n \geq 4$ that if we let $N = n+4$ and

$$A(t) = \sum_{|\alpha| \leq N} \|\Gamma^\alpha \partial u(t, \cdot)\|_{L^2}, \quad 0 \leq t < T$$

then there is a constant $A = A(f, g)$ such that the boot-strap assumption

$$A(t) \leq 2A\epsilon, \quad 0 \leq t \leq T' < T$$

oimplies

$$A(t) \leq A\epsilon/2 + C_A \epsilon \int_0^t \frac{A(s)}{(1+s)^{(n-1)/2}} ds \quad 0 \leq t \leq T'$$

and so by Gronwall's lemma:

$$A(t) \leq A\epsilon/2 + \exp\left[C_A \epsilon \int_0^t \frac{ds}{(1+s)^{(n-1)/2}} \right] \quad 0 \leq t \leq T'$$

when $n \geq 5$, we have that $\int_0^t \frac{ds}{(1+s)^{(n-1)/2}} < \infty$, and so we get a bound $A(t) \leq A\epsilon$ on $[0, T']$ provided that $0 < \epsilon < \epsilon_0$ where ϵ_0 is determined by the condition

$$\exp\left[C_A \epsilon_0 \int_0^| infty \frac{ds}{(1+s)^{(n-1)/2}} \right] = 2$$

showing us that ϵ_0 depends on A, and so on (f, g) (but not on T). Combining this boot-strap argument with the continuity method then gives the *a priori* bound

$$A(t) \leq A\epsilon \quad 0 \leq t < T$$

(finish here, I want to see the counter-example)

6.5.2 The Null Condition And Global Existence For Dimension 3

As we mentioned at the beginning, in general the existence of global smooth solutiosn for small data *fails* in dimension $n = 3$ for equations of type $\Box u = F(\partial u)$. The following was found by F. John, we will not yet show it doesn't have a solution

Example 6.5.1: Non-Existence For Third Dimension
We'll show that every smooth solution of

$$\Box u = (\partial_t u)^2 \quad t \geq 0, \ x \in \mathbb{R}^3$$

with non zero data in $C_c^\infty(\mathbb{R}^3)$ blows up in finite time.

Interestingly, or frustatingly depending on how you see it, there are very similar looking equations that do have a oslution. The following due to Nirenberg:

$$\Box u = (\partial_t u)^2 - \sum_1^3 (\partial_i u)^2 \quad t \geq 0, \ x \in \mathbb{R}^3$$

have global solutions for small data

$$u|_{t=0} = \epsilon f \quad \partial_t u|_{t=0} \epsilon g$$

The key is to notice that

$$v(t,x) = 1 - e^{-u(t,x)}$$

sovles the *linear* Cauchy problem:

$$\Box v = 0 \quad v|_{t=0} 1 - e^{\epsilon f}, \ \partial_t v|_{t=0} \epsilon g e^{-\epsilon f} \tag{6.19}$$

which naturally has a global smooth solution. The inverse transformation of $u \to v$ is

$$u(t,x) = -\log(1 - v(t,x))$$

This is well-defined as long as $|v| < 1$, at which point u solves the Cauchy problem introdued by Nirenberg. To ensure that v is *globally* small, we do

$$\|v(t,\cdot)\|_{L^\infty} < 1 \qquad \text{for all} \qquad t \geq 0$$

we only have to take $\epsilon > 0$ sufficinetly small, depending on f and g. Indeed, remember that in the linear case when $n = 3$ we have the decay estimate

$$\|v(t,\cdot)\|_{L^\infty} \leq \frac{A}{1+t} \qquad \text{for all} \qquad t \geq 0$$

where A is a constant which depends *linearly* on L^∞ norm of $v|_{t=0}$, $\nabla_x v|_{t=0}$ and $\partial_t v|_{t=0}$. In view of the Cauchy problem in equation (6.19), therefor, $A < 1$ if $\epsilon > 0$ is sufficinetly small. Then the transformation $v \to u$ is globally defined, giving a global smooth solution to the original cauchy problem.

These examples suggest that in dimension $n = 3$, the question of global existence of smooth solutions of systems of the type $\Box u = B(\partial u, \partial u)$, where each vector component of B is a bilinear form in ∂u, depends strongly on the algebraic structure of B. More generally, for a system of the form $\Box u = F(\partial u)$, where F vanishes to second order at the origin, it is the quadratic part of F that determines the global regularity properties of the equation. The higher order terms are not important. [Recall from the remark at the end of the previous section that we always have global existence for nonlinearities $F(\partial u)$ which vanish to third order at 0.]

6.5.3 Statement Of Null Condition

We will now consider a system of N equations of the form:

$$\Box u^I = F^I(u\partial u) \qquad (t,x) \in R^{1+3}$$

where the unkonwn u and the given C^∞ functio nF are $\mathbb{R}^n$-valued:

$$u = (u^1, ..., u^N) \qquad F = (F^1, ..., F^N)$$

Definition 6.5.2: Null Vectors

A vector $\xi = (\epsilon_0, \epsilon_1, ..., \epsilon_4)$ is *null* if ξ and $\xi_0^2 = \xi_1^2 + \xi_2^2 + \xi_3^2$. In other words, $|xi$ lies in the ligh cone (or null cone) in Minkowski space $\mathbb{R}^{1+3}$

Definition 6.5.3: Quadratic Part

Let F^I be as in:

$$\Box u^I = F^I(u\partial u) \qquad (t,x) \in R^{1+3}$$

then the *quadratic part* of F^I is

$$F^I_{(2)}(z) = \sum_{|\alpha|=2} \frac{1}{\alpha!} \partial^\alpha F^I(0) z^\alpha$$

where $z \in \mathbb{R}^{N+(n+1)N}$ correponds to $(u, \partial u)$

As we've seen in the previous section, some conditions on the quadratic part of F need to be imposed to insure the existence of a global smooth solution for small data. The relevant priciple is the so called *null conditon of Klainerman and Christodoulou.*

Definition 6.5.4: Null Condition

Let F be as in

$$\Box u^I = F^I(u\partial u) \qquad (t,x) \in R^{1+3}$$

Then we say that F satisfies the *null condition* if

1. F vanishes to second order at the origin:

$$F(0)0 = \qquad DF(0) = 0$$

   Thus, by Taylor's theorem, $F(z) = F_{(2)}(z) + R(z)$ where R is C^∞ and vanishes to third order at 0.

2. The quadratic part of F is of the form

$$F^I_{(2)}(u, \partial u) = \sum_{J,K=1}^{N} \sum_{\mu,\nu=0}^{3} a^{I\mu\nu}_{JK} \partial_\mu u^J \partial_\nu u^K$$

   where the a's are real constants satisfying, fo all $I, J, K = 1, ..., N$

$$\sum_{\mu,\nu=0}^{3} a^{I\mu\nu}_{JK} \xi_\mu \xi_\nu = 0$$

   for all null vectors ξ.

Notice that $F_{(2)}$ is only allowd to depend on ppu, not on u.

Example 6.5.5: Null Condition

Notice that the equation introduced by Nirenberg earlier:

$$\Box u = (\partial_t u)^2 - \sum_1^3 (\partial_i u)^2 \quad t \geq 0,\ x \in \mathbb{R}^3$$

satisfies this condition, while the equation found by F. John:

$$\Box u = (\partial_t u)^2 \quad t \geq 0,\ x \in \mathbb{R}^3$$

does not satisfy the null condition

Lemma 6.5.6: Null Form

If B is a real bilinear form on $\mathbb{R}^n \times \mathbb{R}^4$ such that

$$B(\xi, \xi) = 0$$

for all null vectors $\xi<$ then B is a linear comination, with real coefficients, of the so-called *null forms*:

$$Q_0(\xi, \eta) = \xi_0\eta_0 - \sum_1^3 \xi_i\eta_i$$

$$Q_{\mu\nu}(\xi, \eta) = \xi_\mu\eta_n u - \xi_\nu\eta\mu \qquad 0 \le \mu < \nu \le 3$$

Note that the converse of this lemma is obvious, and so this is the interseting dierction to prove.

***Proof*:**
BY the given conditions, we can deduce that

$$B(\xi, \xi) = \xi^T A\xi = \sum a^{\mu\nu}\xi_\mu\xi\nu$$

where $A = (a^{\mu\nu})$ is a real 4×4 matrix and we condier ξ as a column vector with transpoe $\xi^T = (\xi_0, ..., \xi_3)$. Now decompseo A Into its symetric and skew-symmetric parts:

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2} = A_s + A_a$$

Since $\xi^T A\xi = (\xi^T A\xi)^T = \xi^T A^T\xi$, we see that:

$$\xi^T A_s\xi = \xi^T A\xi = 0$$

for nul ξ. Using this condition with the null vectors:

$$\xi^T = (\pm 1, 1, 0, 0) \quad (\pm 1, 0, 1, 0) \quad (\pm 1, 0, 0, 1)$$

and then with:

$$\xi^T = (\sqrt{2}, 1, 1, 0) \quad (\sqrt{2}, 1, 0, 1) \quad (\sqrt{2}, 0, 1, 1)$$

is not hard to see through some elementary linear algebra that A_s must be of the form:

$$A_s = a^{00}\,\mathrm{diag}(1, -1, -1, -1)$$

which naturally corresponds to Q_0. On the other hand,the skew-symetric part A_a gives a combination of $Q_{|mu\nu}$ in the obvious way. Summing it all up, we have

$$B = a^{00}Q_0 + \sum_{0\le\mu<\nu\le 3} \frac{1}{2}(a^{\mu\nu} - a^{\nu\mu})Q_{\mu\nu}$$

completing the proof.

Corollary 6.5.7: Further Decomposition

Let F be as in equation

$$\Box u^I = F^I(u\partial u) \qquad (t,x) \in R^{1+3}$$

Then F satisfies the null condition if and only if $F^I(u,\partial u)$ is of the form:

$$\sum_{J,K} a^I_{JK} Q_)(\partial u^J, \partial u^K) + \sum_{J,K} \sum_{0\le\mu<\nu\le 3} b^{I_{\mu\nu}}_{JK} Q_{\mu\nu}(\partial u^J, \partial u^K) + R^I(u,\partial u)$$

where the a's and b's are real constants and R^I is smooth and vanishes to third order at 0

Consider:

$$\Box u^I = F^I(u\partial u) \qquad (t,x) \in R^{1+3}$$

with initial data

$$u|_{t=0}\epsilon f \qquad \partial_t u|_{t=0}\epsilon g$$

where $f = (f^1, f^2, ..., f^N)$ and $g = (g^1, g^2, ..., g^N)$ belond to $C^\infty_c(\mathbb{R}^3)$ and $\epsilon > 0>$ Then:

Theorem 6.5.8: Global Existence Under Right Conditions For Low Dimension

Assume that F in the above equation satisfies the null condition. Then there exists $\epsilon_0 = \epsilon_0(f,g) > 0$ such that the above has a smooth global solution provided $\epsilon < \epsilon_0$

6.5.4 Improved Decay

We next tkae a closer look at null forms. In particular, they have better decay proeprties than generic bilinear forms. In order to state the result, we will need some more notation.

(see [5])

6.5.5 An energy inequality and Hörmander's estimate

We need two more results for the main proof. The first is a generalization of the energy inequality

$$\|\partial u(t,\cdot)\|_{L^2} \le \|\partial u(0,\cdot)\|_{L^2} + \int_0^t \|\Box u(s,\cdot)\|_{L^2} ds \tag{6.20}$$

Recall that the original proof was done by noticing that $u_t \Box u$ is a spacetime divergence:

$$u_t \Box u = \mathrm{div}_{t,x}(e_0, e') \tag{6.21}$$

where $e_0 = \frac{1}{2}|\partial u|^2$ and $e' = -u_t \nabla_x u$. Integrating this idenitty for fixed t gives:

$$\int u_t \Box u dx = \frac{d}{dt}\int e_0 dx - \int \mathrm{div}_x(u_t \nabla_x u) dx$$

and the last term vanishes by the divergence theorem, if we assume that u decays sufficenlty fast as $|x| \to \infty$. For the *energy* $E(t) = \int e_0(t,x)dx$ one of then obtains ,after applying the Cauchy-Schwarz inequality to the left ahdn side of the above identity:

$$E'(t) \leq \sqrt{E(t)}\|\Box u(t,\cdot)\|_{L^2}$$

and then the equation at the top of this seciton follows.

We now will want to generalize this method by replacing equation (6.21) with

$$X(\partial)u \cdot \Box u = \mathrm{div}_{t,x}(e_0, e')$$

where $X(\partial)$ is some first order differential operator and (e_0, e') is some spacetime vector invovling u, such that the associated energy is non-negative:

$$E(t) = \int e_0(t,x)dx \geq 0$$

Then, by integrating $X(\partial)u \cdot \Box u = \mathrm{div}_{t,x}(e_0, e')$ one obtains a generalized energy inequality! Set

$$X(\partial)\overline{X} \cdot \partial + 2t \tag{6.22}$$

hwhere as usual ∂ is the spacetime gradient and

$$\overline{X} = (1 + t^2 + |x|^2, 2tx_1, 2tx_2, 2tx_3)$$

Let m be the matrix $\mathrm{diag}(1,-1,-1,-1)$ (i.e. represent the Minkowski metric), and let $\overline{1} = (1,0,0,0)$. It then turns out that equation (6.22) holds with

$$(e_0, e') = X(\partial)u \cdot m(\partial u) - \frac{1}{3}(\partial u)^T m(\partial u)\overline{X} - v^2\overline{1}$$

where we consier ∂u as a column vector for the purposes of matrix multiplication. Integrating equation (6.22) then yeilds us the identity

$$\frac{d}{dt}E(t) = \int X(\partial)u \cdot \Box u dx$$

for the energy $E(t) = \int e_0 dx$, and the right hand side side is bounded by

$$\|(1 + t + |\cdot|)^{-1}X(\partial)u(t,\cdot)\|_{L^2}\|(1 + t + |\cdot|)\Box u(t,\cdot)\|_{L^2}$$

Then we can show that we can bound the left term being multiplied by:

$$\|(1 + t + |\cdot|)^{-1}X(\partial)u(t,\cdot)\|_{L^2} \leq C\sqrt{E(t)}$$

Putting this all together, we obtain:

$$\sqrt{E(t)} \leq C\sqrt{E(0)} + C\int_0^t \|(1 + s + |\cdot|)\Box u(s,\cdot)\|_{L^2} ds$$

Furthermore, one has

$$E(t) \approx \sum_{|\alpha| \leq 1} \|\Gamma^\alpha u(t,\cdot)\|_{L^2}^2$$

Finally, recalling that the commutation relations between $\Box$ and the invariant vector fields ([5]), we finally obtain:

Proposition 6.5.9: Updated Energy Inequality

For any integer $M \geq 0$, there eixsts a constant C such that

$$\sum_{|\alpha|\leq M+1} \|\Gamma^\alpha u(t,\cdot)\|_{L^2}$$
$$\leq$$
$$\sum_{|\alpha|\leq M+1} \|\Gamma^\alpha u(0,\cdot)\|_{L^2} + \sum_{|\alpha|\leq M} \int_0^t \|(1+s+|\cdot|)\Gamma^\alpha \Box u(s,\cdot)\|_{L^2} ds$$

for all $t > 0$ and all $u \in C^\infty([0,\infty)\times\mathbb{R}^3)$ with compact support in x for each t.

***Proof*:**
See [6].

Using these observations, we can state the second result we need (also proven in [6]).

Theorem 6.5.10: Hörmander's Inequality

There exists a C such that if $F \in C^2([0,\infty)\times\mathbb{R}^3)$ and $\Box u = F$ with vanishing initial data at $t = 0<$ then

$$(1+t+|x|) \leq C \sum_{|\alpha|\leq 2} \int_0^t \int_{\mathbb{R}^3} |\gamma^\alpha F(s,y)| \frac{dyds}{1+s+|y|}$$

***Proof*:**
see [6].

6.5.6 Proof of the main Theorem

Since F is assumed to satisfy the null condition ,we konw that the system in the above equaiton satisfies

$$\Box u^I = \sum_{J,k} a^I_{JK} Q_0(\partial u^J, \partial u^K) + \sum_{J,K<\mu,\nu} b^{I\mu\nu}_{JK} Q_{\mu\nu}(\partial u^J, \partial u^K) + R^I(u,\partial u)$$

where R^I vanishes to the third order at 0. We also specify intial data

$$u|_{t=0} = \epsilon f \qquad \partial_t u|_{t=0} \epsilon g$$

For simplicity, we will ignore the higher order term R^I. We shall apply the Hörmander inequality to $\Gamma^\alpha u$ where u solves the above cauchy prbolem, but in order to do thsi we must substract off the solutoin $w_\alpha(w^1_\alpha, ..., w^N_\alpha)$ of the linear Cauchy problem

$$\Box w_\alpha = 0 \quad w_\alpha|_{t=0} = (\Gamma^\alpha u)|_{t=0} \quad \partial_t w_\alpha|_{t=0} = (\partial_t \Gamma^\alpha u)|_{t=0}$$

Thus, $\Gamma^\alpha u - w_\alpha$ has vanishing initial data, so we may apply Hörmander's inequality. But then we also need to esatime $|w_\alpha|$. We start the proof by looking at this estiamte

***Proof*:**
for low dimension non-linear wave equations: We'll first show that if u solves

$$\Box u^I = \sum_{J,k} a^I_{JK} Q_0(\partial u^J, \partial u^K) + \sum_{J,K<\mu,\nu} b^{I_{\mu\nu}}_{JK} Q_{\mu\nu}(\partial u^J, \partial u^K) + R^I(u, \partial u) \tag{6.23}$$

with initial cauchy data

$$u|_{t=0} = \epsilon f \qquad \partial_t u|_{t=0} \epsilon g$$

and w_α solves

$$\Box w_\alpha = 0 \quad w_\alpha|_{t=0} = (\Gamma^\alpha u)|_{t=0} \quad \partial_t w_\alpha|_{t=0} = (\partial_t \Gamma^\alpha u)|_{t=0}$$

then we get

$$|w_\alpha(t,x)| \leq \frac{C_\alpha \epsilon}{1+t} \qquad \forall t \geq 0,\ x \in \mathbb{R}^3$$

where C_α is independnet of t and ϵ.

Proof. Let u_0 sovle $\Box u_0 = 0$ with the above initial data. Then $\Box \Gamma^\alpha u = 0$, and so by the decay estimate for solutiosn of the homogeneous wave equation (remember the result from the chapter)

$$|\Gamma^\alpha u_0(t,x)| \leq \frac{C_\alpha}{1+t} \qquad d\forall t \geq 0,\ x \in \mathbb{R}^3$$

Thus, to get the desired estimate (inequality), it is enoug hto show that

$$|\Gamma^\alpha u_0(t,x) \leq \frac{C_\alpha \epsilon}{1+t} \qquad \forall t \geq 0,\ x \in \mathbb{R}^3$$

Since $\Box(w_\alpha - \Gamma^\alpha u_0) = 0$, this estimate again follows from the decay property we used above, if we just observe that the initial data are $O(\epsilon^2)$. In fact, the data here is

$$\Gamma^\alpha u(0,x) - \Gamma^\alpha u_0(0,x), \qquad \partial_t \Gamma^\alpha u(0,x) - \partial_t \Gamma^\alpha u_0(0,x)$$

To express $\Gamma^\alpha u(0,x)$ in terms of ϵ, f, g, we use equation (6.23) and the cauchy data (whenever there are two or more time derivatives we need to use the equation). If we do the same for $\Gamma^\alpha u(0,x)$ and subract, all terms which are linera in ϵ cancel out, and we are left with terms arising from the nonlinearity in equation (6.23), and which therefore are at least quadratic in ϵ, completing the proof. □

With this result, we now continue onto the main theorem. We break down the proof in 5 steps:

1. Let $0 < T_0 < \infty$ and set $S_{T_0} = [0, T_0) \times \mathbb{R}^3$. Suppose $u \in C^\infty(S_{T_0})$ solves equation (6.23) and the cauchy data (setting $R^I = 0$ for simplicity). We shall prove the existence of $\epsilon_0 > 0<$ independnet of T_0 such that

$$0 < \epsilon < \epsilon_0 \Rightarrow u, \partial u \in L^\infty(S_{T_0})$$

   Once we know this, it follows from local existence that we've done in previous section that the lifespan $T_\epsilon = \infty$

2. The implication given in setp one will follows if we can prove the *a priori* estimate

$$\sum_{|\alpha|\leq k} \|\Gamma^\alpha u(t,\cdot)\|_{L^\infty} \leq \frac{A\epsilon}{1+t} \tag{6.24}$$

for $0 \leq t < T_0 <$ provided $\epsilon < \epsilon_0$. Here A is a constant independent of T_0 and ϵ (A will depend on f and g), and k is a sufficiently large integer $k = 4$ works)

3. The plan to prove the estimate in step 2 is to use the continuity method. Thus, we define

$$E = \{T \in [0, T_0) \mid \text{step 2 holds for all } 0 \leq t \leq T\ \}$$

Clearly $0 \in E$ if we take A sufficinetly large, and E is closed Ifwe can show that E is open in $[0, T_0)$, it will therefore follows that $E = [0, T_0)$, finishing the proof.

To that end, fix $T \in E>$ By continuity of the left hand side of step 2 (note that $\Gamma^\alpha u$ is smooth and compactly supported in x On $[0, T']<$ by Huygen's principle), there certainly exists $T' > T$ such that

$$\sum_{|\alpha|\leq k} \|\Gamma^\alpha u(t,\cdot)\|_{L^\infty} \leq \frac{2A\epsilon}{1+T} \qquad \text{for} \qquad 0 \leq t \leq T' \tag{6.25}$$

The idea now is to use a boot-strap argument to show that we have the stronger estimate shown in step 2 on $[0, T']$. It then will follow that $T' \in E$, proving that E is open. To prove the above estimate implies the estaimte in step 2 when ϵ is sufficinelty small, we first show that the abveo estiamte implies

$$\sum_{|\alpha|\leq k+3} \|\Gamma^\alpha u(t,\cdot)\|_{L^2} \leq C_0(1+t)^{C_1\epsilon} \sum_{|\alpha|\leq k+3} \|\Gamma^\alpha u(0,\cdot)\|_{L^2} \tag{6.26}$$

for $0 \leq t \leq T'$ where C_0 and C_1 are absoltue constants. Then we prove that the above implies the estimate in step 1 if ϵ is sufficeintly small.

4. We know prove that (6.25) $\Rightarrow$ (6.26). Define $A(t)$ to be the left hand side of equation (6.26). Then by the above proposition:

$$A(t) \leq CA(0) + C \sum_{|\alpha\leq k+2} \int_0^t \|(1+s+|\cdot|)\Gamma^\alpha \Box u(s,\cdot)\|_{L^2} ds$$

if we ignore R^I, then $\Box u^I$ is just a linear combination of $Q(\partial u^J, \partial u^K)$ for the null forms Q, and so from [5] we get, taking the higher order derivatives in $QL^2<$ and the lower order in L^∞,

$$A(t) \leq CA(0) + C\int_0^t A(s) \left(\sum_{|\alpha|\leq \frac{k+2}{2}+1} \|\Gamma^\alpha u(s,\cdot)_{L^\infty} \right) ds$$

Since $\frac{k+2}{2} + 1 = \frac{k}{2} + 2$ if $k \geq 4$, the second factor in the integrand is bounded by $\frac{2A\epsilon}{1+s}$ according to equation (6.25), and so we get

$$A(t) \leq CA(0) = C'\epsilon \int_0^t \frac{A(s)}{1+s} ds \qquad 0 \leq t \leq T'$$

Then by Gronwall's lemma:

$$A(t) \leq CA(0) \exp \left[C'\epsilon \int_0^t (1+s)^{-1} ds \right] = CA(0)(1+t)^{C'\epsilon}$$

proving equation (6.26)

5. We prove (6.26) $\Rightarrow$ (6.24), i.e. estimate from step 2. We can naturally choose A so large so that our result from the observation at the begnning of the proof:

$$|w_\alpha(t,x)| \leq \frac{C_\alpha \epsilon}{1+t} \qquad \forall t \geq 0, \ x \in \mathbb{R}^3$$

implies:

$$\sum_{|\alpha| \leq k} \|w_\alpha(t,\cdot)\|_{L^\infty} \leq \frac{A\epsilon/2}{1+t}$$

for all $t \geq 0$> Thus, equation (6.24) follows if we can show that

$$\sum_{|\alpha| \leq k} \|\Gamma^\alpha u(t,\cdot) - w_\alpha(t,\cdot)\|_{L^\infty} \leq \frac{A\epsilon/2}{1+t} \tag{6.27}$$

for $0 \leq t \leq T'$. To prove this we apply Hörmander's inequality, which give

$$(1+t) \sum_{|\alpha| \leq k} \|\Gamma^\alpha u(t,\cdot) - w_\alpha(t,\cdot)\|_{L^\infty} \leq C \sum_{|\beta| \leq 2} \int_0^t \int_{\mathbb{R}^3} \sum_{|\alpha| \leq k} |\Gamma^\beta \Box \Gamma^\alpha u(s,y)| \frac{dyds}{1+s}$$

Using the commutation relation betwen $\Box$ and the invariant vector fields, we may boudn the right hand side by

$$C \sum_{|\alpha| \leq k+2} \int_0^t \int_{\mathbb{R}^3} |\Gamma^\alpha \Box u(s,y)| \frac{dyds}{1+s}$$

and ignore again the R^I it suffices to estimate this with $\Box u$ replaced by $Q(\partial u^J, \partial u^k)$ for ea hof the null fomrs Q. Then if we apply the estimate in [5] we can bound the last expression by

$$C \sum_{|a| \leq k+3} \int_0^t \int_{\mathbb{R}^3} |\gamma^\alpha u(s,y)|^2 \frac{dyds}{(1+s)^2}$$

this equals

$$C \sum_{|a| \leq k+3} \int_0^t |\gamma^\alpha u(s,\cdot)|^2_{L^2} \frac{ds}{(1+s)^2}$$

and using equation (6.26) we bound this by

$$CA(0)^2 \int_0^t (1+s)^{2C_1\epsilon - 2} ds$$

if $2C_1\epsilon < 1$< the integral is uniformally bounded in t, and since $A(0) = O(\epsilon)$< we finally obtain the bound

$$(1+t) \sum_{|\alpha| \leq k} \|\Gamma^\alpha u(t,\cdot) - w_\alpha(t,\cdot)\|_{L^\infty} \leq C\epsilon^2$$

for $0 \leq t \leq T'$, hwere C is an absoltue constant. if $C\epsilon \leq A/2$, then equation (6.27) follows, completing the proof.

7

Future Reading

future reading (following [5]):

1. Strichartz type estimates
2. Applicaiton to Maxell-Klein-Gordon
3. More on well-posedness

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